

The rational Hodge conjecture for CM abelian varieties

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Abstract

We prove the rational Hodge conjecture for complex abelian varieties with complex multiplication: every rational Hodge class on such a variety is a rational linear combination of algebraic cycle classes. As consequences, we obtain the generalized Hodge conjecture for CM abelian varieties, the Tate conjecture for abelian varieties over finite fields, and the Hodge standard conjecture for abelian varieties in arbitrary characteristic.

1 Introduction

For a smooth projective complex variety X , the rational Hodge conjecture asks whether every rational cohomology class of type (p, p) is a rational linear combination of codimension- p algebraic cycle classes. We prove this assertion for abelian varieties with complex multiplication, in every dimension and codimension.

A complex abelian variety A is of *CM type*, or simply *CM*, if $\text{End}(A) \otimes_{\mathbb{Z}} \mathbb{Q}$ contains a commutative semisimple $\mathbb{Q}$ -algebra of dimension $2 \dim A$. We include the point by convention. Finite products and powers of CM abelian varieties are again CM.

Theorem 1.1. *Let A be a complex CM abelian variety. For every integer $p \geq 0$, the Betti cycle-class map*

$$\text{cl}_B : \text{CH}^p(A)_{\mathbb{Q}} \longrightarrow H^{2p}(A, \mathbb{Q}) \cap H^{p,p}(A)$$

is surjective, where $\text{CH}^p(A)_{\mathbb{Q}} = \text{CH}^p(A) \otimes_{\mathbb{Z}} \mathbb{Q}$ and the intersection is taken in $H^{2p}(A, \mathbb{C})$. In particular, the rational Hodge conjecture holds in every codimension on every finite product of complex CM abelian varieties and on every power of such a product.

The theorem also has consequences beyond the algebraicity of individual CM Hodge classes. The generalized Hodge conjecture predicts algebraic supports of the codimension allowed by the Hodge types of a cohomology substructure. We obtain it for CM abelian varieties by applying the theorem to correspondences from auxiliary CM varieties, following Hazama and Abdulali [16, 1]. Two theorems of Milne then give the Tate conjecture for all abelian varieties over finite fields, which asks for algebraic representatives of Galois-invariant ℓ -adic classes, and the Hodge standard conjecture for abelian varieties in arbitrary characteristic, which asserts positivity of the intersection form on primitive algebraic classes [25, 26]. Section 8 states these conclusions with their coefficient and support conventions and explains the separate implications.

1.1 Hodge classes on CM abelian varieties

The codimension-one case of the Hodge conjecture is the Lefschetz $(1, 1)$ theorem. Hodge formulated the broader problem by asking for an extension of this algebraicity criterion to higher-dimensional cycles [18, p. 184]. Cup products of divisor classes therefore give algebraic Hodge classes in higher degree. For an abelian variety, $H^*(A, \mathbb{Q}) = \Lambda^* H^1(A, \mathbb{Q})$, but this description of its cohomology does not make every Hodge class a linear combination of divisor products. The problem is to construct cycles for the remaining classes.

Weil classes provide an important instance of this difficulty. Suppose an imaginary quadratic field K acts on an abelian variety of dimension $2n$, with its two embeddings occurring with equal multiplicity in $H^{1,0}$. Then $\Lambda_K^{2n} H^1(A, \mathbb{Q})$ supplies a two-dimensional rational space of classes of type (n, n) . These determinant classes and their generalization to CM fields are discussed in [10, Proposition 4.4]. Their Hodge type follows from the multiplicity condition; algebraicity requires a geometric construction. Weil classes need not lie in the algebra generated by divisor classes; this occurs already on abelian fourfolds [30, Theorem 0.1]. Recent work of Markman proves algebraicity of the Weil classes on polarized abelian sixfolds of Weil type for an imaginary quadratic field, with polarization discriminant -1 , and deduces the Hodge conjecture for all abelian fourfolds [21, Theorem 1.5.1 and Corollary 1.6.1]. The sixfold theorem assumes an imaginary quadratic action, without requiring the full CM condition of Theorem 1.1.

Deligne proved that every Hodge class on an abelian variety is *absolute Hodge*: after transport to a conjugate variety through de Rham cohomology, it remains a Hodge class [10, Main Theorem 2.11]. This establishes an arithmetic compatibility shared by algebraic cycle classes, but does not by itself produce algebraic cycles. CM varieties play a central role in that argument. Their Hodge groups are tori, and their degree-one cohomology splits, after extending coefficients, into lines indexed by field embeddings. An embedding-line tensor belongs to the scalar extension of the rational Hodge classes exactly when every scalar conjugate has equally many holomorphic and antiholomorphic factors. This is Pohlmann’s balance criterion [33]; see [28, §1.2(c)] and [13, Theorem 3.1].

The CM arguments of Deligne and André reduce these Hodge classes to Weil-type classes on auxiliary CM abelian varieties [10, 2]. More precisely, the formulation in [28, Theorem 1], attributed there to André, expresses a CM Hodge class as a sum of pullbacks of split Weil classes. Here *split* refers to the Hermitian polarization form having a totally isotropic subspace of half its dimension. This reduction identifies classes whose algebraicity would suffice; it does not supply their algebraic representatives in all dimensions.

There is also a reduction to relations involving only four labels. Hazama’s arrangement method spans its rational balance kernel by antipodal pairs and four-label square relations [16, Theorem 7.13]; his Corollary 7.14.1 gives the corresponding codimension-two algebraicity reduction. Gao and Ullmo give equivalent quadratic indicator relations and use them, after adjoining auxiliary CM factors, to generate Hodge-induced period relations [13, Theorems 7.1–7.2]. Their conclusion concerns relations among periods. Our task is to construct algebraic representatives for the corresponding four-factor tensors and to compose those representatives as correspondences. This distinction matters: the reduction uses auxiliary factors and polarization contractions, rather than asserting that low-codimension classes generate the Hodge ring of a fixed power.

1.2 From balanced tensors to surface periods

Choose a Galois CM field E containing the CM fields in $H^1(A, \mathbb{Q})$, and write $G_E = \text{Gal}(E/\mathbb{Q})$ and c for its complex conjugation. An odd sign function $v : G_E \rightarrow \{1, -1\}$, satisfying $v(c\tau) = -v(\tau)$, specifies a weight-one Hodge structure $U(v)$ on the rational vector space E . Its embedding line

$U(v)_\lambda$, on which $a \in E$ acts as $\lambda(a)$, has type $(1, 0)$ or $(0, 1)$ according as $v(\lambda)$ is positive or negative. Section 2 realizes $U(v) = H^1(B(v), \mathbb{Q})$ for a CM abelian variety $B(v)$.

The elementary relation needed in the proof is

$$v_1 + v_2 = v_3 + v_4.$$

It makes every scalar conjugate of the four-factor line

$$U(v_1)_1 \otimes U(v_2)_1 \otimes U(v_3)_c \otimes U(v_4)_c \subset H^4\left(\prod_{i=1}^4 B(v_i), \overline{\mathbb{Q}}\right)$$

have type $(2, 2)$. We prove that this line lies in the scalar extension of the rational cycle span. Section 2 then assembles arbitrary balanced tensors from these relations: two-row switches connect lists with the same sum, and algebraic polarization pairings contract their intermediate factors. Transfer back to the original abelian variety and descent of coefficients give Theorem 1.1.

The geometric construction of a four-factor class takes place on a surface. Suppose there are rational Hodge maps $h_i : U(v_i) \rightarrow H^1(S, \mathbb{Q})$, for a smooth projective surface S , and vectors α_i on the four selected embedding lines such that

$$\int_S h_1(\alpha_1) \smile h_2(\alpha_2) \smile h_3(\alpha_3) \smile h_4(\alpha_4) \neq 0.$$

Degree-one Hodge maps are induced, up to rational multiples, by morphisms $S \rightarrow B(v_i)$. The image of S in their product therefore gives an algebraic cycle whose pairing with the desired line is nonzero. Section 3 shows how four algebraic divisor kernels convert this functional into a vector on that exact line. The remaining problem is thus to find four degree-one classes with both a nonzero surface period and the prescribed rational CM sources.

1.3 Theta forms and their CM sources

Section 4 constructs the period on a compact arithmetic quotient of the complex two-ball. Its unitary group has signature $(2, 1)$ at one distinguished embedding and is definite at all the other real places. After the elementary normalization in Section 2, all four lines $U(v_i)_1$ are holomorphic at that embedding. To each type the construction assigns a family of holomorphic theta one-forms. Their values span a cotangent fiber, so two such families supply a wedge that is a nonzero holomorphic two-form.

To compare the two sides of the four-factor relation, we realize each pair of types by an orthogonal decomposition of the same Hermitian plane. The balance relation says that the two decompositions have the same local sign multiplicities. Weak approximation produces a rational second decomposition whose compact-place data are as close as desired to those of the first. At the distinguished embedding, the total period data depend only on the plane and remain unchanged.

A rank-two theta tensor packages the wedges into one total two-form. We keep its finite theta input and the compact quotient fixed while varying the compact-place data. The total form varies continuously, so its pairing with the initial nonzero form stays close to the nonzero self-pairing. At the two rational endpoints it is a finite sum of wedges of the required one-forms. Factoring these endpoints and passing to a common finite cover therefore gives a nonzero mixed period with two holomorphic and two antiholomorphic factors.

The construction uses Weil's rational theta covariance and the unitary splittings of the Weil representation [37, 19, 5]. Kudla and Millson developed differential-form theta kernels in this

unitary setting [20, Introduction and Theorem 3]. The projection turning the rank-two tensor into wedges has a close predecessor in the pure-cocycle product calculation of Bergeron, Millson, and Mœglin [3, Lemmas 5.7 and 5.11]. We derive the required tensor identity in coordinates and use the fixed-input variation to obtain the mixed period.

These are initially complex one-form classes. The surface criterion also requires that they come from specified rational CM Hodge structures. Sections 5 and 6 establish this arithmetic input. First we realize the ball quotients in a projective moduli space and construct algebraic Hecke endomorphisms of the product of its components' Albanese varieties. Counting ordinary isogeny neighbors gives a polynomial relation between these operators and Frobenius. We then compute the Hecke parameters of the theta forms. The Frobenius relation and the degree-one Weil weight force slope zero or one on the relevant Hecke summands, with the choice determined by the prescribed CM signs.

The Hecke projectors preserve the Hodge filtration, so weak admissibility turns these extreme slopes into pure Hodge types. Applying the calculation to every scalar conjugate identifies the complete CM type of each summand. Scalar descent then expresses every relevant theta class as a complex linear combination of images of the selected CM line under rational Hodge maps. Section 7 puts the finite families on one projective surface, expands the nonzero period, and applies the surface criterion. This completes the four-factor construction and hence the proof of the main theorem.

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2 CM tensors and four-factor relations

This section reduces Theorem 1.1 to the algebraicity of a specified family of tensors in the fourth cohomology of products of four CM abelian varieties. The reduction changes the CM types of individual factors while preserving their total Hodge type; its intermediate factors may therefore be absent from the original variety. We first record how to work with individual embedding lines and recover rationality after extending coefficients. We then construct the auxiliary CM varieties and assemble arbitrary balanced tensors from four-factor relations by algebraic correspondences.

2.1 Scalar coefficients and degree-one maps

For a smooth projective complex variety X , write

$$\mathrm{Hdg}^{2p}(X) = H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X), \quad \mathcal{Z}^{2p}(X) = \mathrm{im}(\mathrm{CH}^p(X)_{\mathbb{Q}} \longrightarrow H^{2p}(X, \mathbb{Q})).$$

Cohomology is Betti cohomology unless another theory is specified. We fix $\overline{\mathbb{Q}} \subset \mathbb{C}$, and write $T_k = T \otimes_{\mathbb{Q}} k$ for a rational vector space T . A class with coefficients in $\overline{\mathbb{Q}}$ or $\mathbb{C}$ is called *algebraic* if it belongs to the corresponding scalar extension of $\mathcal{Z}^{2p}(X)$. Scalar conjugation therefore preserves algebraicity on the same complex variety: it fixes each rational cycle class and changes only its coefficient. We use the compatibility of cycle classes with exterior products, cup products, pullback between smooth varieties, and proper pushforward.

Our Tate convention is $V(r)^{a,b} = V^{a+r,b+r}$, so $V(r)$ has weight $\mathrm{wt}(V) - 2r$.

Lemma 2.1 (Scalar descent). *Let T be a finite-dimensional rational vector space and $P \subset T_{\mathbb{C}}$ a complex-linear subspace. If $t \in T_{\overline{\mathbb{Q}}}$ and $\sigma t \in P$ for every $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, then*

$$t \in (T \cap P) \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}.$$

The same assertion applies to the subspace of maps of any fixed Hodge type in a Hom space.

Proof. Choose a finite Galois extension $K/\mathbb{Q}$ containing the coefficients of t . For $a \in K$, the vector

$$T_a = \mathrm{Tr}_{K/\mathbb{Q}}(at) = \sum_{\sigma \in \mathrm{Gal}(K/\mathbb{Q})} \sigma(a)\sigma(t)$$

is rational. Each automorphism of K extends to $\overline{\mathbb{Q}}$, so every summand belongs to P ; hence $T_a \in T \cap P$. If $b_1, \dots, b_r$ is a rational basis of K and $b_1^*, \dots, b_r^*$ is its trace-dual basis, write $t = \sum_j t_j b_j$ with $t_j \in T$. Then $T_{b_j^*} = t_j$, which proves the assertion. The condition that a complex-linear map have a fixed Hodge type is complex-linear, so the same argument applies in a Hom space. $\square$

We will also use

$$(\mathcal{Z}^{2p}(X) \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}) \cap H^{2p}(X, \mathbb{Q}) = \mathcal{Z}^{2p}(X). \quad (2.1)$$

Indeed, extend a rational basis of $\mathcal{Z}^{2p}(X)$ to one of $H^{2p}(X, \mathbb{Q})$ and compare coefficients.

Lemma 2.2 (Realization in degree one). *For complex abelian varieties A and B , pullback induces an isomorphism*

$$\mathrm{Hom}(A, B) \otimes_{\mathbb{Z}} \mathbb{Q} \simeq \mathrm{Hom}_{\mathrm{Hdg}}(H^1(B, \mathbb{Q}), H^1(A, \mathbb{Q})).$$

If S is a smooth connected projective complex variety and $h : H^1(B, \mathbb{Q}) \rightarrow H^1(S, \mathbb{Q})$ is a rational Hodge map, then there are an integer $m \geq 1$ and a morphism $f : S \rightarrow B$ such that $f^ = mh$ on $H^1(B, \mathbb{Q})$.*

Proof. Dualizing a rational Hodge map gives a rational map on homology commuting with the complex structures. Multiplication by a positive integer makes it carry the integral lattice of A into that of B . It consequently induces a holomorphic homomorphism of complex tori $A^{\mathrm{an}} \rightarrow B^{\mathrm{an}}$, which is algebraic. Conversely, an algebraic homomorphism induces the indicated Hodge pullback. This proves the first assertion; equivalently it is the dual of the rational form of Riemann's theorem [27, Proposition 6.5, Theorems 6.7–6.8, and Corollary 6.9].

The Albanese map of S induces an isomorphism $H^1(\mathrm{Alb} S, \mathbb{Q}) \simeq H^1(S, \mathbb{Q})$. Apply the first assertion to the map with target $H^1(\mathrm{Alb} S, \mathbb{Q})$, clear a denominator, and compose the resulting homomorphism $\mathrm{Alb} S \rightarrow B$ with an Albanese map. $\square$

The Lefschetz $(1, 1)$ theorem says that $\mathrm{Hdg}^2(X) = \mathcal{Z}^2(X)$. In conjunction with Lemma 2.1, it has the following useful consequence: if $t \in H^2(X, \overline{\mathbb{Q}})$ is of type $(1, 1)$ in every scalar conjugate, then t is algebraic. This observation will supply all two-factor tensors used below.

2.2 Rank-one CM structures

For a polarizable rational Hodge structure H of weight one, let $J(H)$ be the smallest rational algebraic subgroup of $\mathrm{GL}(H)$ containing the Hodge circle. The circle acts by z on $H^{1,0}$ and by $\bar{z}$ on $H^{0,1}$, for $|z| = 1$. It preserves an alternating polarization. Its connectedness shows that it lies in the rational identity component of $J(H)$; minimality therefore makes $J(H)$ connected.

Lemma 2.3. *For a complex abelian variety A , the following conditions are equivalent:*

- (i) $J(H^1(A, \mathbb{Q}))$ is a torus.
- (ii) $H^1(A, \mathbb{Q})$ is a direct sum of rational Hodge structures, each of rank one over a CM field acting by Hodge endomorphisms.
- (iii) $\mathrm{End}(A) \otimes_{\mathbb{Z}} \mathbb{Q}$ contains a commutative semisimple algebra of rational dimension $2 \dim A$.

Proof. The zero-dimensional case is immediate. Put $H = H^1(A, \mathbb{Q})$, and first assume that $J(H)$ is a torus. Its associative envelope $E_0 \subset \mathrm{End}_{\mathbb{Q}}(H)$ is a commutative semisimple algebra: after splitting the torus, its elements act diagonally by scalars on the weight spaces, so the envelope is a subalgebra of a product of fields and is reduced. This description descends to $\mathbb{Q}$.

Let $*$ be the polarization adjoint. It preserves E_0 , since it sends every symplectic torus element to its inverse. Every element of E_0 commutes with the complex structure I supplied by the Hodge circle. Consequently $*$ is also the adjoint for the positive definite real form obtained by inserting I in the polarization. In particular,

$$\mathrm{Tr}_{H_{\mathbb{R}}}(a^*a) > 0 \quad (0 \neq a \in E_0 \otimes_{\mathbb{Q}} \mathbb{R}).$$

The involution cannot exchange two primitive factors of the real finite étale algebra $E_0 \otimes \mathbb{R}$: for the idempotent e of one factor, this would give $e^*e = 0$. On a complex factor it cannot be the identity, since for its idempotent e one would have $(ie)^*(ie) = -e$, of negative trace. Thus it acts as complex conjugation on each complex factor. There are no real factors. On the nonzero module of such a factor the circle would act by real scalars, and a continuous homomorphism from the circle to $\mathbb{R}^{\times}$ is trivial; this contradicts the weight-one characters.

It follows that each rational field factor K of E_0 is CM. Indeed, it has only complex archimedean factors, and its involution satisfies $\tau(a^*) = \overline{\tau(a)}$ at every embedding. Its fixed field is therefore totally real and has index two. On the corresponding K -module the circle acts by $K \otimes \mathbb{R}$ -scalars. Splitting that module into K -lines thus gives rational Hodge substructures. This proves (ii), without discarding multiplicities.

The separate scalar actions on the field lines in (ii) form a commutative semisimple Hodge-endomorphism algebra of dimension $\dim_{\mathbb{Q}} H$. Lemma 2.2 realizes it in $\mathrm{End}(A) \otimes \mathbb{Q}$, proving (iii). Conversely, write an algebra as in (iii) as $\prod_j K_j$. Its faithful action on H has a positive rank m_j over each K_j , and

$$\dim_{\mathbb{Q}} H = \sum_j m_j [K_j : \mathbb{Q}] = \sum_j [K_j : \mathbb{Q}].$$

Hence every $m_j = 1$. The centralizer of this action in $\mathrm{GL}(H)$ is the torus $\prod_j \mathrm{Res}_{K_j/\mathbb{Q}} \mathbf{G}_m$. It contains the Hodge circle because the algebra acts by Hodge endomorphisms, proving (i). $\square$

These equivalences give the rank-one description of the CM varieties defined in the introduction. A finite product of CM abelian varieties is CM: the circle on the direct sum of their degree-one cohomologies lies in the product of their tori, and its connected rational envelope is a subtorus.

Choose a Galois CM field $E \subset \overline{\mathbb{Q}}$, and put

$$F = E^+, \quad G_E = \text{Gal}(E/\mathbb{Q}), \quad s = [F : \mathbb{Q}].$$

Its complex conjugation $c \in G_E$ is central. One such E can contain any finite collection of CM fields and all their conjugates. To see this, take the compositum of all those conjugates. For a CM field K , every embedding $\rho : K \rightarrow \mathbb{C}$ satisfies $\rho c_K = c_{\mathbb{C}} \rho$. These identities show that complex conjugation on the Galois compositum commutes with its Galois group. Every conjugate of an element fixed by it is real, so the fixed field is totally real. The compositum is therefore a Galois CM field.

An *odd sign function* on G_E is a function $v : G_E \rightarrow \{1, -1\}$ satisfying $v(c\tau) = -v(\tau)$. Give the rational vector space E a weight-one Hodge structure $U(v)$ by declaring its embedding line

$$U(v)_\lambda = \{u \in U(v)_{\overline{\mathbb{Q}}} : au = \lambda(a)u \text{ for every } a \in E\} \quad (\lambda \in G_E)$$

to have type $(1, 0)$ if $v(\lambda) = 1$, and type $(0, 1)$ otherwise. Oddness makes these declarations compatible with the real structure.

This Hodge structure is polarizable. Choose $0 \neq \delta \in E$ with $c(\delta) = -\delta$. Multiplying δ by an element of $F^\times$, weak approximation gives an element $\xi \in E^\times$ such that $c(\xi) = -\xi$ and

$$\text{sgn}(\text{Im } \tau(\xi)) = -v(\tau)$$

at one representative τ of each conjugate pair. The form

$$\psi_v(x, y) = \text{Tr}_{E/\mathbb{Q}}(\xi x c(y))$$

is alternating and nondegenerate. It pairs only conjugate embedding lines. On a vector u in the sum of the $(1, 0)$ lines, $i\psi_v(u, \bar{u})$ is a sum of positive multiples of the squared absolute values of its embedding coordinates; hence it is positive for $u \neq 0$. This is the weight-one polarization condition. After scaling, the form is integral on a lattice. The Riemann criterion, applied to the dual homological Hodge structure, gives a CM abelian variety $B(v)$ of dimension s with

$$U(v) = H^1(B(v), \mathbb{Q}). \quad (2.2)$$

This construction permits every odd sign function, including imprimitive CM types; it does not require a principal polarization. The polarization and Riemann-criterion argument also applies to any CM field, without a Galois hypothesis, and any CM type, that is, any choice of one embedding from each conjugate pair. The Galois field E provides a common indexing set for the types used below.

We will need the exact action of scalar conjugation on labels. If $\sigma \in \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ restricts to $g \in G_E$ and $u \in U(v)_\lambda$, rationality of the E -action gives

$$a \sigma(u) = \sigma(au) = \sigma(\lambda(a))\sigma(u) = (g\lambda)(a)\sigma(u).$$

Consequently

$$\sigma U(v)_\lambda = U(v)_{g\lambda}. \quad (2.3)$$

In particular the complete type signs of $U(v)_1$ and $U(v)_c$ after g are $v(g)$ and $-v(g)$, respectively. Here and below the commutation of c with g takes place in G_E ; no centrality assertion is made in $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$.

The same convention applies to a line in a rank-one module over a CM subfield of E . Choose $\lambda \in G_E$ extending the embedding that labels the line. Its *complete type function* takes the value $+1$ at $g \in G_E$ if the line labeled by the restriction of $g\lambda$ has type $(1, 0)$, and -1 if it has type $(0, 1)$. It is odd. Since the embedding-line decomposition is defined over E , this is the type of the scalar-conjugate line for every automorphism of $\overline{\mathbb{Q}}$ restricting to g .

Lemma 2.4 (Transfer of an embedding line). *Let H and H' be direct sums of rank-one weight-one Hodge modules over CM fields contained in E . Choose an embedding line $L \subset H_{\overline{\mathbb{Q}}}$ and an embedding line $M \subset H'_{\overline{\mathbb{Q}}}$ with the same complete type function. Extend any $\overline{\mathbb{Q}}$ -linear map $T : L \rightarrow M$ by zero on all other source embedding lines. Then*

$$T \in \text{Hom}_{\text{Hdg}}(H, H') \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}.$$

When H and H' are degree-one cohomologies of abelian varieties, this map is therefore an algebraic-coefficient linear combination of pullbacks of abelian morphisms, up to rational denominators.

Proof. Scalar conjugation transports the entire source and target decompositions. For each σ , the only possibly nonzero component of σT maps σL to σM , whose types agree by hypothesis. All other components are zero. Thus σT is a Hodge map on the whole source. Apply Lemma 2.1 in $\text{Hom}_{\mathbb{Q}}(H, H')$, followed, when applicable, by Lemma 2.2. $\square$

The map T in this lemma need not be invariant under a stabilizer of either line. This matters when a field is a proper subfield of E , or when a rational field module occurs more than once. The proof treats every scalar conjugate of the zero-extended map and retains the repeated rational summands separately.

2.3 Reduction to four factors

The four-factor assertion that remains to be proved is the following.

Theorem 2.5 (Four-factor switch). *Let E be a Galois CM field and let $v_1, v_2, v_3, v_4 : G_E \rightarrow \{1, -1\}$ be odd sign functions satisfying $v_1 + v_2 = v_3 + v_4$. Under the Künneth inclusion, the following line is algebraic:*

$$U(v_1)_1 \otimes U(v_2)_1 \otimes U(v_3)_c \otimes U(v_4)_c \subset H^4 \left(\prod_{i=1}^4 B(v_i), \overline{\mathbb{Q}} \right) \quad (2.4)$$

This line is already of Hodge type $(2, 2)$ in every scalar conjugate. Indeed, after an automorphism restricting to $g \in G_E$, its four type signs have sum

$$v_1(g) + v_2(g) - v_3(g) - v_4(g) = 0. \quad (2.5)$$

The theorem asks for algebraicity, which does not follow from this balance alone. We leave its proof to the subsequent construction; the present section proves that it is sufficient.

Hazama's coordinate-arrangement calculation rationally spans the kernel of its Hodge matrix by antipodal pairs and elementary four-label relations [16, Theorem 7.13]. His Corollary 7.14.1 states a corresponding codimension-two algebraicity reduction. We prove the particular tensor reduction needed here directly, including the transfers to auxiliary abelian factors and the algebraic contractions that assemble the relations.

Proposition 2.6. *If Theorem 2.5 holds for every Galois CM field and every quadruple of odd sign functions satisfying its relation, then Theorem 1.1 holds.*

Proof. Let A be a CM abelian variety and let $h \in \text{Hdg}^{2p}(A)$. The cases $p = 0$ and $p = 1$ follow from the unit class and the Lefschetz $(1, 1)$ theorem, respectively. For higher codimension, we first replace a Hodge monomial by a tensor on auxiliary CM factors. We then connect its two lists of types by four-factor switches and compose the resulting algebraic tensors.

By Lemma 2.3, choose a rational decomposition of $H = H^1(A, \mathbb{Q})$ into rank-one CM-field modules, retaining separate copies of repeated summands. Choose a Galois CM field E containing

all the fields and their conjugates. Over E , H is the direct sum of its embedding lines. Since cup product gives $H^*(A, \mathbb{Q}) = \Lambda^* H$, the exterior monomials in these lines give a direct-sum decomposition of $H^*(A, \overline{\mathbb{Q}})$ into one-dimensional lines [27, Proposition 6.4]. Every such line is contained in one Hodge bidegree.

Scalar conjugation permutes the monomial lines, with only the possible sign from reordering exterior factors. Because h is rational, a nonzero component on one monomial line gives a nonzero component on every line in its scalar-Galois orbit. All these lines must have type (p, p) . The decomposition is defined over E , so the components of h have algebraic Betti coefficients.

Fix a supported monomial and list the complete type functions of its $2p$ factors as $z_1, \dots, z_{2p}$. Each z_i is odd. In the conjugate indexed by g , the number of $(1, 0)$ factors minus the number of $(0, 1)$ factors is $\sum_i z_i(g)$. Its balance is therefore exactly

$$\sum_{i=1}^{2p} z_i = 0.$$

For $1 \leq i \leq p$, set $x_i = z_i$ and $y_i = -z_{p+i}$. Then $\sum_i x_i = \sum_i y_i$. The first p factors have the complete types of $U(x_i)_1$, and the last p have those of $U(y_i)_c$. For any ordered list $a = (a_1, \dots, a_p)$ of odd sign functions, put $A_a = \prod_{i=1}^p B(a_i)$; for two such lists put

$$\mathcal{L}_{a,b} = \bigotimes_{i=1}^p U(a_i)_1 \otimes \bigotimes_{i=1}^p U(b_i)_c \subset H^{2p}(A_a \times A_b, \overline{\mathbb{Q}}).$$

Repeated factors in these products are kept as separate slots. Lemma 2.4 reduces the chosen monomial to the algebraicity of $\mathcal{L}_{x,y}$. We return to A after composing the transition tensors.

Regard each list as a matrix with p rows and one column for each representative of $G_E / \langle c \rangle$. At any column, equality of the two sums is equality of the numbers of plus signs. Among the rows where the current x -entry differs from the desired y -entry, there are equally many mismatches of the forms $(+1, -1)$ and $(-1, +1)$. Pair these two sets of rows and swap the two current entries in every pair, making the opposite swap at the conjugate coordinate. This corrects the column, leaves the other columns unchanged, and preserves oddness of every row. Successively correcting the columns gives a finite sequence

$$x = x^{(0)}, x^{(1)}, \dots, x^{(N)} = y$$

in which consecutive lists differ by one such two-row swap.

At a single swap the old rows a, b and new rows a', b' satisfy $a + b = a' + b'$ at every coordinate. Theorem 2.5 supplies the pure algebraic tensor

$$U(a)_1 \otimes U(b)_1 \otimes U(a')_c \otimes U(b')_c.$$

Every unchanged row is joined to its new copy by $U(v)_1 \otimes U(v)_c$, an all-conjugate $(1, 1)$ line and hence an algebraic two-factor tensor. Pull these classes to their distinct old and new slots and multiply. This gives a nonzero algebraic generator of $\mathcal{L}_{x^{(j)}, x^{(j+1)}}$. If the original lists already agree, the two-factor tensors alone generate $\mathcal{L}_{x,x}$. It remains to compose consecutive transition tensors without losing their algebraicity or nonvanishing.

For three ordered lists a, b, e , suppose that $\mathcal{L}_{a,b}$ and $\mathcal{L}_{b,e}$ have pure algebraic generators with algebraic Betti coefficients. Write them as

$$\begin{aligned} t_{ab} &= \alpha_1 \otimes \dots \otimes \alpha_p \otimes \beta_1 \otimes \dots \otimes \beta_p, \\ t_{be} &= \gamma_1 \otimes \dots \otimes \gamma_p \otimes \delta_1 \otimes \dots \otimes \delta_p, \end{aligned}$$

where $\beta_i \in U(b_i)_c$ and $\gamma_i \in U(b_i)_1$. Identify the middle copies and choose an ample divisor class ℓ_i on each $B(b_i)$. With r_{ab}, r_{be}, r_{ae} denoting the projections from $A_a \times A_b \times A_e$, and with the ℓ_i also denoting their pullbacks, cup product and pushforward give

$$(r_{ae})_* \left(r_{ab}^* t_{ab} \smile r_{be}^* t_{be} \smile \prod_{i=1}^p \ell_i^{s-1} \right) = (-1)^{p(p-1)/2} \left(\prod_{i=1}^p q_i \right) t_{ae}, \quad (2.6)$$

$$q_i = \int_{B(b_i)} \beta_i \wedge \gamma_i \wedge \ell_i^{s-1},$$

where $t_{ae} = \alpha_1 \otimes \cdots \otimes \alpha_p \otimes \delta_1 \otimes \cdots \otimes \delta_p$. The sign is obtained by regrouping $\beta_1 \cdots \beta_p \gamma_1 \cdots \gamma_p$ into $\beta_1 \gamma_1 \cdots \beta_p \gamma_p$ in the indicated slot order.

Each q_i is nonzero. Complex conjugation exchanges the two middle embedding lines, so γ_i is a nonzero scalar multiple of $\bar{\beta}_i$. Hodge–Riemann positivity for the ample class makes the pairing with $\bar{\beta}_i$ nonzero, whether β_i is holomorphic or antiholomorphic. This requires no compatibility between ℓ_i and E . Moreover $q_i \in \overline{\mathbb{Q}}^\times$, since its vectors have algebraic Betti coefficients and ℓ_i is rational. The codimension on the left of Equation (2.6) is

$$p + p + p(s-1) - ps = p.$$

Thus it is an algebraic class of the required degree, and division by the nonzero algebraic coefficient shows that the endpoint line is algebraic. Iterating along the sequence of lists proves algebraicity of $\mathcal{L}_{x,y}$.

Finally choose the line transfers in the direction from the auxiliary U 's to $H^1(A, \mathbb{Q})$. Each is an algebraic-coefficient linear combination of rational pullbacks along maps from A to the corresponding $B(v)$. Apply their exterior product on A^{2p} and pull back along the diagonal $A \rightarrow A^{2p}$. The result is the cup product of the selected original factors. They occupy distinct embedding or multiplicity lines in the original exterior monomial, so this cup product is nonzero. We have proved that every supported monomial line is algebraic. Summing the components of h and using Equation (2.1) proves the rational assertion.

Finite products are covered because they are themselves CM, as observed after Lemma 2.3. $\square$

The proof uses tensors on auxiliary products and contracts their intermediate slots before returning to A . It does not assert generation of the Hodge ring on a fixed power by cup products of classes in codimensions one and two.

Lemma 2.7 (Choice of the distinguished embedding). *It suffices to prove Theorem 2.5 when $s \geq 2$ and $v_i(1) = 1$ for all four functions.*

Proof. If v_1 and v_2 are nowhere simultaneously positive, oddness also rules out simultaneous negativity. Hence $v_2 = -v_1$. The relation gives $v_4 = -v_3$, and the first two and the last two slots of Equation (2.4) are separately all-conjugate $(1, 1)$ pairs.

Otherwise choose $g \in G_E$ at which $v_1(g) = v_2(g) = 1$. The relation makes all four signs positive there. Choose a scalar automorphism extending g . It sends the labels in Equation (2.4) to g, g, gc, gc . The complete type function of the line labeled g is the right translate

$$v_i^g(\tau) = v_i(\tau g).$$

It is odd, satisfies $v_i^g(1) = 1$, and the four translated functions satisfy the same relation. Centrality of c in G_E makes the complete type of the gc -line equal to $-v_i^g$. The transfers of Lemma 2.4 therefore

carry the normalized switch for the v_i^g 's to the lines labeled g, g, gc, gc on the original factors. Applying the inverse scalar automorphism gives Equation (2.4).

When $s = 1$, there is only one odd sign function positive at the identity, so the normalized tensor is a product of ordinary pairs. $\square$

For the remaining construction we work in the normalized case, so all four lines $U(v_i)_1$ are holomorphic. The next section isolates a surface-period condition sufficient to prove the switch.

3 A surface criterion for the switch

To prove Theorem 2.5, we will construct four degree-one classes on one surface with a nonzero cup product. The criterion below explains why this is sufficient. The four maps give an algebraic surface class that tests the switch line nontrivially; four divisor classes then convert that test into a vector on the line itself.

Proposition 3.1 (Surface criterion). *Let E be a Galois CM field with conjugation c , and let $v_1, v_2, v_3, v_4 : G_E \rightarrow \{1, -1\}$ be odd sign functions satisfying $v_1 + v_2 = v_3 + v_4$. Put $\lambda = (1, 1, c, c)$. Suppose that S is a smooth connected projective complex surface and that there are rational Hodge maps*

$$h_i : U(v_i) \longrightarrow H^1(S, \mathbb{Q}), \quad 1 \leq i \leq 4,$$

and nonzero vectors $\alpha_i \in U(v_i)_{\lambda_i} \subset U(v_i)_{\overline{\mathbb{Q}}}$ such that

$$\int_S h_1(\alpha_1) \smile h_2(\alpha_2) \smile h_3(\alpha_3) \smile h_4(\alpha_4) \neq 0. \quad (3.1)$$

Here each h_i is extended to $\overline{\mathbb{Q}}$ -coefficients. Then the line in Equation (2.4) is algebraic.

Proof. By Lemma 2.2, there are positive integers m_i and morphisms $f_i : S \rightarrow B(v_i)$ with $f_i^* = m_i h_i$. Therefore

$$k := \int_S f_1^* \alpha_1 \smile f_2^* \alpha_2 \smile f_3^* \alpha_3 \smile f_4^* \alpha_4 \in \overline{\mathbb{Q}}^\times. \quad (3.2)$$

Nonvanishing follows from Equation (3.1), since the new value is multiplied by $\prod_i m_i$. The value is algebraic because the pullbacks, cup product, and evaluation on $[S]$ are rational operations on Betti cohomology and the α_i have algebraic Betti coefficients.

Put $A = \prod_i B(v_i)$, $n = \dim A = 4s$, and $f = (f_i) : S \rightarrow A$. Let

$$z = f_*[S] \in \mathrm{CH}^{n-2}(A), \quad Z = \mathrm{cl}(z) \in \mathcal{Z}^{2n-4}(A).$$

For $t_\alpha = \alpha_1 \otimes \cdots \otimes \alpha_4$ in the four- H^1 Künneth summand of $H^4(A, \overline{\mathbb{Q}})$, the projection formula gives

$$\int_A Z \smile t_\alpha = k. \quad (3.3)$$

Thus Z gives a nonzero functional on the switch line; it is not itself a vector on that line. In particular $f(S)$ cannot have dimension less than two, since then the proper pushforward of the two-cycle $[S]$ and the displayed pairing would be zero.

For each i , take a copy $B(v_i)'$ with its transported E -action, and choose a nonzero vector $\beta_i \in U(v_i)'_{c\lambda_i}$ with algebraic Betti coefficients. Under the Künneth inclusion define the pure class

$$D_i = \alpha_i \otimes \beta_i \in H^2(B(v_i) \times B(v_i)', \overline{\mathbb{Q}}).$$

If $\sigma \in \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ restricts to $g \in G_E$, then Equation (2.3) puts its two factors on the lines labeled

$$g\lambda_i, \quad gc\lambda_i = cg\lambda_i.$$

These have opposite Hodge types. This uses centrality of c in G_E only, and remains valid when the scalar coordinates of α_i and β_i lie outside E . Every scalar conjugate of D_i therefore has type $(1, 1)$. Scalar descent and the Lefschetz $(1, 1)$ theorem give

$$D_i \in \mathcal{Z}^2(B(v_i) \times B(v_i)') \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}. \quad (3.4)$$

The entire cohomology class D_i is algebraic here. It was defined in the pure $H^1 \otimes H^1$ summand before applying descent, so there is no need to project an arbitrary divisor class onto that summand.

Set $A' = \prod_i B(v_i)'$. Let $p : A \times A' \rightarrow A$ and $p' : A \times A' \rightarrow A'$ be the projections, and let $q_i : A \times A' \rightarrow B(v_i) \times B(v_i)'$ select the i -th old and new slots. Write

$$K = \prod_{i=1}^4 q_i^* D_i, \quad t_\beta = \beta_1 \otimes \cdots \otimes \beta_4.$$

Order the old factors first and the new factors second in the Künneth identification for $A \times A'$. Moving the new degree-one factor of D_i past the later old degree-one factors makes $3 + 2 + 1 = 6$ sign changes. Hence

$$K = (-1)^6 t_\alpha \otimes t_\beta = t_\alpha \otimes t_\beta.$$

In particular K has degree four in the old factors and degree four in the new factors. Integration over A consequently gives the exact formula

$$p'_*(p^*Z \smile K) = \left(\int_A Z \smile t_\alpha \right) t_\beta = k t_\beta. \quad (3.5)$$

Any other Künneth component of Z contributes zero to this integration. The codimension of the left side is $(n - 2) + 4 - n = 2$, equivalently its cohomological degree is $(2n - 4) + 8 - 2n = 4$.

By Equation (3.4), the left side of Equation (3.5) is algebraic with $\overline{\mathbb{Q}}$ -coefficients. Since $k \in \overline{\mathbb{Q}}^\times$, division by k proves algebraicity of the line generated by t_β , whose labels are $(c, c, 1, 1)$. Ordinary scalar complex conjugation fixes the rational cycle span on the same A' and changes these labels to $(1, 1, c, c)$. Identifying the copied factors with the original ones gives the asserted line on A . $\square$

The theta construction produces complex classes, and the arithmetic argument will place them in spans of rational Hodge images. The following consequence applies to those classes.

Corollary 3.2 (Surface criterion for complex spans). *Let $E, v_1, \dots, v_4$, and $\lambda = (1, 1, c, c)$ be as in Proposition 3.1, and let S be a smooth connected projective complex surface. Suppose that, for each $1 \leq i \leq 4$, a class $\eta_i \in H^1(S, \mathbb{C})$ belongs to*

$$\text{span}_{\mathbb{C}}\{h(U(v_i)_{\lambda_i}) : h \in \text{Hom}_{\text{Hdg}}(U(v_i), H^1(S, \mathbb{Q}))\}$$

and that

$$\int_S \eta_1 \smile \eta_2 \smile \eta_3 \smile \eta_4 \neq 0.$$

Then the line in Equation (2.4) is algebraic.

Proof. Fix a nonzero algebraic Betti generator of each selected line. Each η_i is a finite complex linear combination of its images under rational Hodge maps. Expanding the nonzero cup product selects a nonzero term of the form in Equation (3.1). Thus Proposition 3.1 applies. This argument uses only finite sums; all four maps have the same target $H^1(S, \mathbb{Q})$. $\square$

4 Theta differentials and a nonzero mixed period

The surface criterion of Proposition 3.1 asks for a nonzero period made from four one-forms on one surface. We construct the analytic part of that input using theta series attached to auxiliary Hermitian spaces of dimensions one and two, called multiplicity spaces. For a multiplicity line, a component of the first Taylor tensor is a holomorphic one-form. Varying the finite theta input makes its values span a cotangent fiber, so two such constructions give a nonzero wedge. For their product input, the second Taylor tensor for the orthogonal sum of the two lines has a double-alternating projection that is a nonzero multiple of that wedge. This total two-form allows us to change the two multiplicity lines while keeping their sum and its finite theta input fixed.

For the two sign pairs in the normalized switch, we will obtain total two-forms ξ, ξ' on one fixed compact ball quotient X . On a common principal cover $\pi : X' \rightarrow X$, they have finite expansions into holomorphic one-forms:

$$\pi^* \xi = \sum_r a_r \Theta_1(\phi_{1,r}) \wedge \Theta_2(\phi_{2,r}), \quad \pi^* \xi' = \sum_s b_s \Theta_3(\phi_{3,s}) \wedge \Theta_4(\phi_{4,s})$$

such that

$$\int_{X'} \pi^* \xi \wedge \overline{\pi^* \xi'} \neq 0. \quad (4.1)$$

The comparison of ξ' with ξ keeps the rational symplectic space, the rational special-unitary group, the total finite theta input, and the compact quotient fixed. Only afterward will we factor the rational endpoints and pass to X' , on which the individual one-forms in both sums descend. At this stage they give complex cohomology classes. Section 5 will provide projective principal levels, and Section 6 will identify the rational CM source maps.

4.1 The ball and the sign data

Let $E \subset \overline{\mathbb{Q}}$ be a Galois CM field, put $F = E^+$, and assume $s = [F : \mathbb{Q}] \geq 2$. Write $G_E = \text{Gal}(E/\mathbb{Q})$ and let $c \in G_E$ be complex conjugation. Fix a set $\Sigma \subset G_E$ of representatives modulo c , including the distinguished embedding $\mu = 1$. The construction for a single multiplicity line takes an odd sign function $v : G_E \rightarrow \{1, -1\}$ with $v(1) = 1$. We index its signs by

$$d(\tau) = v(\tau^{-1}), \quad d(c\tau) = -d(\tau), \quad d(\mu) = 1. \quad (4.2)$$

Thus $d(g^{-1}) = v(g)$, the indexing needed by the arithmetic calculation. For the normalized switch, fix four such functions with $d_1 + d_2 = d_3 + d_4$. We first construct the one-form for an arbitrary one of them, and then compare the pairs (d_1, d_2) and (d_3, d_4) . All Hermitian forms in this section are linear in their first argument.

Choose an E/F -Hermitian space V of E -dimension three, of signature $(2, 1)$ at μ and positive definite at every other real place of F . For example, take a diagonal form with two totally positive coefficients and a third coefficient negative at μ and positive elsewhere; weak approximation in F supplies the last coefficient. Put

$$G = \text{SU}(V), \quad D = \{ L \subset V_\mu : \dim_{\mathbb{C}} L = 1, h_V|_L < 0 \}.$$

The group G is defined over F , and D is the complex two-ball. We also write L for the tautological negative line bundle over D , and set $Q = \underline{V}_\mu / L$. The Grassmannian tangent calculation gives

$$T_D = \text{Hom}(L, Q), \quad \Omega_D^1 = L \otimes Q^*. \quad (4.3)$$

Fix an $\mathcal{O}_E$ -lattice $\Lambda \subset V$; all principal congruence levels below are matrix levels for this lattice. Their quotients of D are compact. To see the relevant anisotropy, an isotropic vector defined over E would still be isotropic at a definite real completion, which is impossible. Equivalently, G has no nontrivial F -split torus: a nonzero weight space of such a torus in V would be isotropic, since invariance of h_V forces $h_V(x, x) = 0$ for a vector of nontrivial split weight. Thus G is F -anisotropic. The arithmetic lattice and compactness theorems therefore make its arithmetic subgroups cocompact lattices in $G(F \otimes_{\mathbb{Q}} \mathbb{R})$ [6]; see also the formulation in [27, Theorem 3.3 and Proposition 3.5]. Here

$$G(F \otimes_{\mathbb{Q}} \mathbb{R}) \simeq \mathrm{SU}(2, 1) \times \prod_{\tau \neq \mu} \mathrm{SU}(3).$$

Projection past the compact factors preserves discreteness and cocompactness. Indeed, a sequence in an arithmetic subgroup whose active projections approach the identity has, after taking a subsequence, convergent compact components; differences would then approach the identity in the full discrete subgroup. A sufficiently deep principal matrix level is torsion-free. Its action on D is free, because a discrete subgroup has finite intersection with a point stabilizer. Hence its quotient is a smooth connected compact Kähler surface. Later we will impose a principal level for which the reference moduli construction also proves projectivity.

4.2 Positive Lagrangians and differential bundles

Fix $\delta \in E^\times$ with $c(\delta) = -\delta$. For a nondegenerate E/F -Hermitian space P of E -dimension $n \in \{1, 2\}$, let

$$\mathbb{V}_P = \mathrm{Res}_{E/F}(V \otimes_E P).$$

It is an F -space of dimension $6n$. The product of the two Hermitian forms is a well-defined Hermitian form on $V \otimes_E P$, and we use the F -valued form

$$\psi_P(v \otimes p, v' \otimes p') = \mathrm{Tr}_{E/F}(\delta h_V(v, v') h_P(p, p')). \quad (4.4)$$

Interchanging the arguments conjugates the Hermitian product and changes the sign of δ , so ψ_P is alternating. It is nondegenerate: the tensor-product Hermitian form is nondegenerate, and the trace pairing on E/F is nondegenerate. More explicitly, if the Hermitian pairing of $x \neq 0$ with some y is $b \neq 0$, then its pairing with ey , for $e \in E$, is $c(e)b$; trace duality supplies e for which $\mathrm{Tr}_{E/F}(\delta c(e)b) \neq 0$. Thus $(\mathbb{V}_P, \psi_P)$ is a symplectic space over F . Its underlying rational vector space carries the alternating form $\mathrm{Tr}_{F/\mathbb{Q}} \psi_P$; this is the restriction of scalars used in complexifications over $\mathbb{Q}$ below. Symplectic coordinates and transformations will still be taken over F .

First take $P = W_i$ to be a Hermitian line associated with a sign function d_i as in (4.2). Complexifying the trace pairing identifies the $c\tau$ summand complex-linearly with the dual of the τ summand. Define a subbundle $\mathcal{F}_i \subset \mathbb{V}_P \otimes_{\mathbb{Q}} \mathbb{C}$ over D at the active pair of embeddings by

$$(\mathcal{F}_i)_\mu = L \otimes (W_i)_\mu, \quad (\mathcal{F}_i)_{c\mu} = Q^* \otimes (W_i)_\mu^*. \quad (4.5)$$

Here $Q^* = \mathrm{ann}(L) \subset V_\mu^*$, embedded in the conjugate summand by the Hermitian pairing. It is the holomorphic annihilator bundle, not an antiholomorphically varying orthogonal complement in V_μ . At $\tau \in \Sigma \setminus \{\mu\}$, take the full $c\tau$ summand when $d_i(\tau) = 1$, and the full τ summand when $d_i(\tau) = -1$; take zero on the other summand. For a rational orthogonal sum $P = W_1 \perp W_2$, take the direct sum of these prescriptions and write $\mathcal{F}_P = \mathcal{F}_1 \oplus \mathcal{F}_2$.

Lemma 4.1. *The Hermitian forms on the lines W_i can be chosen over E/F so that $\mathcal{F}_P$ is a holomorphic positive Lagrangian for ψ_P , equivariantly for $G(F \otimes_{\mathbb{Q}} \mathbb{R})$. Its determinant and an equivariant square root have the form*

$$\det \mathcal{F}_P \simeq L^{2n} \otimes C_P, \quad (\det \mathcal{F}_P)^{1/2} \simeq L^n \otimes C_P^{1/2}, \quad (4.6)$$

where C_P is a constant complex line on which $G(F \otimes_{\mathbb{Q}} \mathbb{R})$ acts trivially.

Proof. Write $\tau(\delta) = it_\tau$, where $t_\tau \in \mathbb{R}^\times$. At the real place represented by $\tau \in \Sigma$, write $\psi_{P,\tau}$ for the complex bilinear extension of the real-place form. Our positivity convention is

$$\psi_{P,\tau}(u, \bar{u})/i > 0 \quad (0 \neq u \in (\mathcal{F}_P)_\tau \oplus (\mathcal{F}_P)_{c\tau}).$$

On the τ summand its sign is the sign of $t_\tau h_V h_P$; on the $c\tau$ summand the sign is the opposite. At μ , choose the coefficient of each line form to have sign $-\text{sgn}(t_\mu)$. The negative line L on the μ side and the dual of the positive plane on the $c\mu$ side then both have positive sign. At a definite place $\tau \neq \mu$, choose the line coefficient to have sign

$$-d_i(\tau) \text{sgn}(t_\tau).$$

The chosen full summand has positive sign in either case. These finitely many strict sign conditions on an element of F are simultaneously realizable by weak approximation.

At the active pair the two subspaces are annihilators under the complexified pairing; at every other pair one side is full and the other zero. Thus $\mathcal{F}_P$ is isotropic of half the total dimension. The active annihilator description proves holomorphy, and the compact-place summands are constant over D . Equivariance follows from the action on L and its annihilator at μ , and from preservation of the full embedding summands at the compact places.

Finally

$$\det Q^* = L \otimes \det(V_\mu)^*.$$

For general P the active determinant is therefore

$$\det(L \otimes P_\mu) \otimes \det(Q^* \otimes P_\mu^*) \simeq L^{2n} \otimes (\det(V_\mu)^*)^{\otimes n} \otimes \det(P_\mu)^*.$$

All factors other than L^{2n} , including the determinants at the compact places, are constant lines. Their special-unitary actions are trivial, because they involve powers of determinants of V_τ or V_τ^* , while G acts trivially on P . This proves the first isomorphism in (4.6); choosing a square root of the constant line gives the second. $\square$

We choose nonzero complex vectors in these constant lines when identifying tensors with differential forms. Such choices only rescale those forms by nonzero constants. No rational trivialization is asserted. Also, the constant lines are trivial for $G = \text{SU}(V)$; they need not be trivial under the larger unitary group.

The determinant calculation gives the differential targets for the theta tensors. For a multiplicity line W_i , the relevant target is

$$\begin{aligned} (\det \mathcal{F}_i)^{1/2} \otimes (\mathcal{F}_i)_{c\mu} &\simeq L \otimes Q^* \otimes (\text{constant line}) \\ &\simeq \Omega_D^1 \otimes (\text{constant line}). \end{aligned} \quad (4.7)$$

For $W = W_1 \perp W_2$, define the double-alternating map

$$\text{alt}_2 : \text{Sym}^2(Q^* \otimes W_\mu^*) \longrightarrow \Lambda^2 Q^* \otimes \Lambda^2 W_\mu^*$$

by

$$\text{alt}_2((a \otimes u)(b \otimes v)) = (a \wedge b) \otimes (u \wedge v).$$

Here the product on the left is multiplication in the symmetric algebra. Exchanging its two factors changes both exterior signs, so the map is well-defined. Including the half-determinant line gives

$$(\det \mathcal{F}_W)^{1/2} \otimes \Lambda^2 Q^* \otimes \Lambda^2 W_\mu^* \simeq K_D \otimes (\text{constant line}), \quad (4.8)$$

because $K_D = \det \Omega_D^1 = L^2 \otimes \Lambda^2 Q^*$. The first theta tensor will give the one-form in (4.7); the double-alternating part of the second will give the two-form in (4.8).

The reason to combine the two lines is geometric. At the active pair, their total positive Lagrangian has the pieces

$$L \otimes W_\mu, \quad Q^* \otimes W_\mu^*.$$

These depend only on W_μ , independently of its decomposition into lines. At a compact place, let W_τ^+ and W_τ^- be the sums of the multiplicity lines with respective signs $+1$ and -1 under d_i . The total positive pieces there are

$$V_\tau \otimes W_\tau^-, \quad V_\tau^* \otimes (W_\tau^+)^*;$$

the second dual is embedded as $\text{ann}(W_\tau^-) \subset W_\tau^*$. The relation $d_1 + d_2 = d_3 + d_4$ says that the two sign pairs have the same multiset at each place. Thus the original local lines can be relabeled with signs d_3, d_4 without changing these total pieces. Weak approximation will produce a rational decomposition with the new signs and nearby compact-place data, while leaving the active total Lagrangian unchanged. We will apply this approximation in Subsection 4.7 to one fixed theta input. To compare the resulting forms on a compact quotient, we first establish their covariance and congruence descent.

4.3 Rational theta covariance

Choose symplectic coordinates over F on $\mathbb{V}_P$, written $(p, q) \in F^a \oplus F^a$, where $a = 3n$, with

$$\psi_P((p, q), (p', q')) = p^t q' - q^t p'.$$

The Schrödinger position coordinates are q . A positive Lagrangian at a real place has a unique graph frame

$$e_Z = \begin{pmatrix} Z \\ I \end{pmatrix}, \quad Z = Z^t, \quad \text{Im } Z > 0.$$

Indeed, it cannot meet the p -axis nontrivially, because $\psi_P(u, \bar{u})$ vanishes there. Projection to the q -axis is therefore an isomorphism. Isotropy says $Z^t = Z$, and, for a graph vector,

$$\psi_{P,\tau}(e_Z u, \overline{e_Z u})/i = 2u^t(\text{Im } Z)\bar{u},$$

which proves the last condition. The positive Lagrangians above thus give a period map $D \rightarrow \prod_{\tau \in \Sigma} \mathfrak{H}_a$.

Let $\phi \in \mathcal{S}(\mathbb{A}_{F,f}^a)$ be a finite-adelic Schwartz–Bruhat function. For $Z = (Z_\tau)_\tau \in \prod_\tau \mathfrak{H}_a$ and $z = (z_\tau)_\tau \in \bigoplus_\tau \mathbb{C}^a$, put

$$\vartheta_\phi(Z, z) = \sum_{x \in F^a} \phi(x) \exp \left(\pi i \sum_{\tau \in \Sigma} x_\tau^t Z_\tau x_\tau + 2\pi i \sum_{\tau \in \Sigma} x_\tau^t z_\tau \right). \quad (4.9)$$

The rational transformation law for this series includes an exponential quadratic in z , which mixes its ordinary Taylor coefficients. The following completion cancels that factor and gives coefficients with a tensor transformation law. Define

$$\begin{aligned}\widehat{\vartheta}_\phi(Z, z) &= \exp\left(\frac{\pi}{2} \sum_{\tau \in \Sigma} z_\tau^t (\operatorname{Im} Z_\tau)^{-1} z_\tau\right) \vartheta_\phi(Z, z), \\ \widehat{\vartheta}_\phi(Z, z) &= \sum_{k \geq 0} \langle T_{k, \phi}(Z), z^{\otimes k} \rangle.\end{aligned}\tag{4.10}$$

Thus $T_{k, \phi}$ includes the factor $1/k!$ in the k -th derivative. We use multiplication in the symmetric algebra for these homogeneous coefficients.

The series and every fixed order of its derivatives converge uniformly on compact subsets of the Siegel and z domains. Indeed, the rational points in the support of ϕ lie in finitely many translates of a full lattice in $(F \otimes_{\mathbb{Q}} \mathbb{R})^a$. On a compact set the imaginary parts have a common positive lower bound, and the summands, after a fixed number of differentiations, are bounded by a polynomial in $\|x\|$ times

$$C \exp(-c\|x\|^2 + C'\|x\|).$$

The lattice sum of this bound converges. The completion and its derivatives obey the same bounds on compact sets. The same conclusion holds after any fixed number of derivatives in smooth parameters on a compact subset of a smooth parameter domain: there the period map and all derivatives of it needed for that order are bounded, and its imaginary parts have a common positive lower bound. Compactness of the image of a period map alone, without bounds on its parameter derivatives, would not imply this last assertion.

We next specify the rational Weil-representation input, including the cover being used. The circle extension is obtained from the metaplectic double cover by extending its central $\{1, -1\}$ to the group of unit complex scalars. Take the global additive character of $\mathbb{A}_F/F$ whose infinite component is $t \mapsto \exp(2\pi i \operatorname{Tr}_{F/\mathbb{Q}} t)$. Weil's rational section in the adelic circle extension leaves the summation functional

$$\lambda(\Phi) = \sum_{x \in F^a} \Phi(x)$$

invariant. In characteristic zero his pseudosymplectic group is the symplectic group, and his adelic twofold subgroup contains this rational section [37, Ch. III, nos. 31 and 41, Eq. (39) and Theorem 6; Ch. IV, no. 45, pp. 200–201]. Consequently, for a rational symplectic transformation we may choose an infinite double-cover lift and its paired finite lift whose adelic product is the rational lift. A change of the infinite sign requires the same change of the finite sign. These paired lifts give rational covariance; they are not an assertion that the symplectic double cover splits on an arbitrary adelic unitary group.

Lemma 4.2. *Let $g \in \operatorname{Sp}(\mathbb{V}_p)(F)$. At each real place write*

$$g = \begin{pmatrix} A & B \\ C & D_0 \end{pmatrix}, \quad J = CZ + D_0, \quad gZ = (AZ + B)J^{-1}.$$

Choose an infinite metaplectic lift with factor $j(g, Z)$ satisfying $j(g, Z)^2 = \prod_{\tau \in \Sigma} \det J_\tau$, and let $\tilde{g}_f$ be its paired finite lift. Then

$$\vartheta_{\omega_f(\tilde{g}_f)\phi}(gZ, J^{-t}z) = j(g, Z) \exp\left(\pi i \sum_{\tau \in \Sigma} z_\tau^t J_\tau^{-1} C_\tau z_\tau\right) \vartheta_\phi(Z, z).\tag{4.11}$$

The completed coefficients satisfy

$$T_{k, \omega_f(\tilde{g}_f)\phi}(gZ) = j(g, Z) \operatorname{Sym}^k(J) T_{k, \phi}(Z). \quad (4.12)$$

They are thus tensors valued in

$$(\det \mathcal{F}_P)^{1/2} \otimes \operatorname{Sym}^k \mathcal{F}_P.$$

In particular the positive Lagrangian, not its dual, occurs in the symmetric factor. The tensor $T_{1, \phi}$ is holomorphic.

Proof. We first check the infinite formula, keeping track of its signs. At one real place write

$$f_{Z, z}(x) = \exp(\pi i x^t Z x + 2\pi i x^t z), \quad K(g, Z) = J^{-1}C.$$

The matrix $K(g, Z)$ is symmetric; this follows from $JC^t = CJ^t$, a symplectic identity. The formula to prove is

$$\omega_\infty(\tilde{g})f_{Z, z} = j(g, Z)^{-1} \exp(-\pi i z^t K(g, Z)z) f_{gZ, J^{-t}z}. \quad (4.13)$$

For symmetric B , the upper unipotent $n(B) = \begin{pmatrix} I & B \\ 0 & I \end{pmatrix}$ acts by multiplication by $\exp(\pi i x^t B x)$, which gives (4.13) with $j = 1$. For $m(A) = \operatorname{diag}(A, A^{-t})$, the chosen lift acts by

$$\omega_\infty(\tilde{m}(A))b(x) = j(m(A))^{-1}b(A^t x), \quad j(m(A))^2 = (\det A)^{-1}.$$

For $\det A > 0$ the positive choice has $j(m(A))^{-1} = |\det A|^{1/2}$. For negative determinant the metaplectic phase is retained in j . Since $m(A)Z = AZA^t$ and $J^{-t}z = Az$, this is the required formula.

For $w = \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}$, choose the lift

$$b \longmapsto e^{-\pi i a/4} \int_{\mathbb{R}^a} b(y) e^{-2\pi i x^t y} dy.$$

The Gaussian integral is

$$\int_{\mathbb{R}^a} f_{Z, z}(y) e^{-2\pi i x^t y} dy = \det(-iZ)^{-1/2} \exp(-\pi i (x - z)^t Z^{-1} (x - z)),$$

where the square root of $\det(-iZ)$ equals 1 at $Z = iI$. Taking

$$j(w, Z) = e^{\pi i a/4} \det(-iZ)^{1/2}$$

gives $j(w, Z)^2 = \det Z$ and proves (4.13) for w .

These generators give the formula for the whole real symplectic group. To check that the displayed exponent also composes correctly, let $J_2 = J(g_2, Z)$ and $J_1 = J(g_1, g_2 Z)$. The symplectic identities give

$$J(g_1 g_2, Z) = J_1 J_2, \quad K(g_1 g_2, Z) = K(g_2, Z) + J_2^{-1} K(g_1, g_2 Z) J_2^{-t}.$$

For the second identity, use $A_2 - (g_2 Z)C_2 = J_2^{-t}$ to write $C_{12} = C_1 J_2^{-t} + J_1 C_2$, and multiply by $(J_1 J_2)^{-1}$. Together with the metaplectic group law for j , this is exactly the composition law in (4.13). Multiplying the formula over the real places gives its infinite adelic version.

Apply rational invariance of λ to $\phi \otimes \prod_\tau f_{Z_\tau, z_\tau}$. By (4.13), the transformed sum is

$$j(g, Z)^{-1} \exp\left(-\pi i \sum_\tau z_\tau^t J_\tau^{-1} C_\tau z_\tau\right) \vartheta_{\omega_f(\tilde{g}_f)\phi}(gZ, J^{-t}z).$$

Equating this with the original sum gives (4.11). In particular the finite action on the left is $\omega_f(\tilde{g}_f)$, with no inverse. If one instead keeps the finite input on the left fixed, the input on the right is acted on by $\omega_f(\tilde{g}_f)^{-1}$.

For the completion, put $Y = \text{Im } Z$ and $Y' = \text{Im}(gZ)$ at one place. From $Y' = \bar{J}^{-t} Y J^{-1}$, and then $\bar{J}^t = J^t - 2iY C^t$, we obtain

$$J^{-1}(Y')^{-1} J^{-t} = Y^{-1} - 2iJ^{-1}C. \quad (4.14)$$

Here $C^t J^{-t} = J^{-1}C$ by symmetry. Thus the completion factor at $(gZ, J^{-t}z)$ differs from that at (Z, z) by $\exp(-\pi i z^t J^{-1} C z)$. This cancels the exponential in (4.11), so

$$\hat{\vartheta}_{\omega_f(\tilde{g}_f)\phi}(gZ, J^{-t}z) = j(g, Z) \hat{\vartheta}_\phi(Z, z).$$

Comparison of homogeneous terms yields (4.12). Finally,

$$g e_Z = e_{gZ} J.$$

Hence J is the transformation matrix for vectors in the positive Lagrangian; this proves the asserted tensor target and its orientation. The completion has no linear term in z , so $T_{1,\phi}$ is the ordinary holomorphic theta gradient. For comparison, the first-order law agrees with the classical theta-gradient formula in [29, §9.1]. $\square$

For a multiplicity line W_i , the $c\mu$ -projection of $T_{1,\phi}$ therefore lies in the cotangent target (4.7). Rational covariance alone does not supply a congruence subgroup fixing a given finite input. Subsection 4.4 obtains such stabilizers from an honest smooth finite-adelic action.

Remark 4.3. We will use the lemma to change rational Schrödinger coordinates. Suppose a rational symplectic matrix S takes adapted coordinate vectors to the reference coordinate vectors. Then $Z_{\text{ref}} = S Z_{\text{ad}}$, and the input in the adapted coordinates is

$$\phi_{\text{ad}} = \omega_f(\tilde{S}_f)^{-1} \phi_{\text{ref}}.$$

Equation (4.12) compares the resulting tensors. Changing the sign of the paired lift changes both the infinite factor and the finite input sign, so their comparison is unchanged. For coordinates adapted to a rational symplectic direct sum, the period matrix is block diagonal and the completed theta series of a product input is the product of the two completed series.

4.4 Unitary splittings and congruence descent

The preceding lemma uses paired lifts of rational symplectic elements. An honest action on finite inputs, and therefore a congruence stabilizer for each input, requires a different statement. We use the unitary splitting theorem in the following precise form. For a quadratic extension E/F , a Hermitian space H of dimension m , a skew-Hermitian space K of dimension n , and a global additive character, choose unitary Hecke characters χ_H, χ_K of E with

$$\chi_H|_{\mathbb{A}_F^\times/F^\times} = \epsilon_{E/F}^m, \quad \chi_K|_{\mathbb{A}_F^\times/F^\times} = \epsilon_{E/F}^n.$$

The corresponding unitary actions lift to the *circle-scalar extension* of the adelic symplectic Weil representation. The global lift is assembled from local lifts and agrees with the canonical rational symplectic section on rational unitary points. Its finite part acts smoothly on finite Schwartz functions. Changing the splitting characters twists the unitary actions by characters composed with their determinants. Here $\epsilon_{E/F}$ is the quadratic idele class character.

The local splitting is due to Kudla [19]. We use the circle-extension formulation and its local and adelic compatibility in [5, §§3.1–3.1.1 and 4.1–4.1.1]. In the present CM setting, adelic automorphy and the determinant-twist rule are also recorded in [3, §7.2 and Remark 7.3]. Throughout this invocation the extension is the circle pushout: we do not assert that the double cover splits on every unitary group. We will compare this circle lift with an actual double lift only after restriction to $G = \mathrm{SU}(V)$. The split-place normalization below will also enter the local Hecke calculation of Subsection 6.1.

Lemma 4.4. *For the symplectic space $(\mathbb{V}_P, \psi_P)$, restriction of the unitary splitting to $G(\mathbb{A}_F)$ yields an honest finite-adelic Weil action $\omega_{P,f}$. It is compatible with the rational covariance in Lemma 4.2 and is independent of the splitting characters. At infinity its half-determinant action is the natural action on L^n in (4.6).*

At a split finite place, choose as Schrödinger positions for a multiplicity line one standard three-dimensional embedding summand. The resulting SL_3 -action is inverse linear substitution. More generally, the normalized linear action associated with a transformation h of the position space is

$$(\rho(h)b)(x) = |\det h|^{-1/2} b(h^{-1}x). \quad (4.15)$$

A bare metaplectic lift outside SL_3 is the normalized operator $\rho(h)$ times its lift scalar; that scalar must be retained in any use of such a lift.

Every finite list of inputs is fixed under $\omega_{P,f}$ by all sufficiently deep principal lattice congruence subgroups.

Proof. First we check that the required splitting characters exist. For the idele class groups C_F and C_E , the natural map $C_F \rightarrow C_E$ is injective: if an F -idele becomes a principal E -idele, its generator is fixed by c and hence belongs to F . This map is also proper. The absolute idele norm on C_E restricts to the square of the norm on C_F , whose norm-one kernel is compact. A compact subset of C_E has norms in a compact subinterval of $\mathbb{R}_{>0}$; its preimage is therefore a closed subset of a compact norm interval times the compact norm-one kernel of C_F . Thus C_F is a closed locally compact subgroup of C_E . The character-extension theorem for a closed subgroup of a locally compact abelian group extends each unitary character $\epsilon_{E/F}^r$ from C_F to C_E [12, §4.3, Corollary 4.42]. Apply this with $r = m, n$.

The splitting theorem applies to our form by replacing h_P with the skew-Hermitian form δh_P . With both forms linear in the first variable, the trace product in [5, §§2.1 and 3.1] is exactly (4.4). We take the same global additive character as in Lemma 4.2. The determinant twists from changing the characters vanish on $G = \mathrm{SU}(V)$. The product of the finite local splittings is therefore an honest action $\omega_{P,f}$ on $G(\mathbb{A}_{F,f})$, independent of those characters.

We compare its infinite circle lift with the double lift needed for the differential bundles. At the active maximal compact subgroup $S(\mathrm{U}(2) \times \mathrm{U}(1))$, let ℓ be the character on the negative line. The determinant of the positive plane is ℓ^{-1} , so the determinant character on $\mathcal{F}_P$ is ℓ^{2n} . At every other real place the determinant character is trivial on $\mathrm{SU}(3)$. On the maximal compact $\mathrm{U}(a)$ of each real symplectic factor, the metaplectic double cover is the determinant-square-root cover. For the product of the real factors its square is the product of those determinant characters. Its pullback to the maximal compact subgroup of $G(F \otimes_{\mathbb{Q}} \mathbb{R})$ therefore splits with square-root character ℓ^n .

A connected real reductive group retracts onto its maximal compact subgroup. The preceding determinant computation says that every fundamental-group loop of $G(F \otimes_{\mathbb{Q}} \mathbb{R})$ has even determinant winding in the symplectic group. The lift from the universal covering group consequently descends to a group lift of $G(F \otimes_{\mathbb{Q}} \mathbb{R})$ into the double cover. Its half-determinant action is the one on L^n in (4.6). The original circle lift and this double lift, pushed out to the circle extension, differ

by a continuous character of $G(F \otimes_{\mathbb{Q}} \mathbb{R})$. This group is connected and semisimple. The differential of such a character is zero, and connectedness makes the character trivial. The two lifts therefore agree. The rational compatibility of the global splitting now pairs $\omega_{P,f}(\gamma)$ with this infinite lift for every $\gamma \in G(F)$, proving the claimed rational covariance.

At a split finite place the two embedding summands are dual stable Lagrangians. If h is the action on the position summand, the symplectic matrix is $\text{diag}(h^{-t}, h)$. The Schrödinger linear formula gives (4.15), up to the scalar of the chosen bare lift. On $\text{SL}_3(F_v)$, the normalized substitution lift and the unitary lift have a character as their ratio. This character is trivial because $\text{SL}_3(F_v)$ is perfect: elementary matrices generate it, and for distinct i, j, k ,

$$e_{ij}(u) = [e_{ik}(u), e_{kj}(1)].$$

The determinant factor is also 1 there. This proves the split-place assertion without removing scalars from lifts outside SL_3 .

Finally the finite Weil action is smooth, so finitely many Schwartz functions have a common open stabilizer. Intersect it with the fixed lattice stabilizer in $G(\mathbb{A}_{F,f})$. Principal lattice congruence neighborhoods form a basis at the identity in that stabilizer, so a sufficiently deep principal one lies in the intersection. This proves the last assertion. $\square$

Restrict the tensors of Lemma 4.2 to the period map from D . If a finite list of inputs is fixed by an open compact $K_f \subset G(\mathbb{A}_{F,f})$, the lemma just proved and (4.6) give their bundle covariance for $\Gamma = G(F) \cap K_f$. Passing to a torsion-free principal subgroup makes all of them descend to one compact quotient. This uses stabilizers in the smooth finite-adelic representation; it makes no claim that an arbitrary finite-index subgroup of a ball lattice is congruence.

4.5 Theta one-forms span cotangent fibers

For a line W_i , project $T_{1,\phi}$ to the $c\mu$ summand of the positive Lagrangian. Using our fixed complex trivialization of the constant line in (4.7), denote the resulting one-form on D by $\Theta_i(\phi)$.

Proposition 4.5. *For every finite input ϕ , the form $\Theta_i(\phi)$ is holomorphic on D and descends to every sufficiently deep principal congruence quotient. At any fixed point of D , the values $\Theta_i(\phi)$, as ϕ varies over finite Schwartz inputs, span the cotangent fiber. Finitely many actual inputs already span that fiber and descend at a common sufficiently deep principal level.*

Proof. Holomorphy follows from that of $T_{1,\phi}$, the holomorphic embedding projection, and (4.3). Descent follows from Lemma 4.4.

Fix a point $x_* \in D$ and its period point Z . Here $a = 3$. For $0 \neq x_0 \in \mathcal{O}_F^a$ and positive integers k , take

$$\phi_k = \mathbf{1}_{x_0 + k\hat{\mathcal{O}}_F^a}.$$

Its rational support is exactly $x_0 + k\mathcal{O}_F^a$. The first Taylor tensor is the ordinary gradient, so in the chosen graph frame

$$T_{1,\phi_k}(Z) = 2\pi i \sum_{x \in x_0 + k\mathcal{O}_F^a} (x_\tau)_{\tau \in \Sigma} \exp\left(\pi i \sum_{\tau} x_\tau^t Z_\tau x_\tau\right).$$

Under the Minkowski embedding $\mathcal{O}_F^a$ is a full lattice. If $\rho > 0$ is the minimum norm of a nonzero lattice vector, then, for $m \neq 0$ in this lattice and $k \geq 2\|x_0\|/\rho$,

$$\|x_0 + km\| \geq k\|m\|/2.$$

The Gaussian bound used above, with any polynomial factor absorbed into a weaker Gaussian, bounds the tail by a constant times a convergent lattice sum of the form $\sum_{m \neq 0} \exp(-ck^2 \|m\|^2)$. That sum tends to zero. The same argument is uniform for Z in a compact neighborhood. We have proved

$$T_{1,\phi_k}(Z) \longrightarrow 2\pi i (x_{0,\tau})_{\tau \in \Sigma} \exp\left(\pi i \sum_{\tau} x_{0,\tau}^t Z_{\tau} x_{0,\tau}\right). \quad (4.16)$$

The scalar exponential is nonzero. As x_0 varies, the vectors $(x_{0,\tau})_{\tau}$ span the real position space, and their complex span is the entire first-tensor fiber in this frame. The complex span of the actual values $T_{1,\phi}(Z)$ is a subspace of a finite-dimensional space and hence is closed. It contains all the limits in (4.16), so it is the full fiber. Projection onto $(\det \mathcal{F}_i)^{1/2} \otimes (\mathcal{F}_i)_{c\mu}$ is surjective; therefore the projected values span $\Omega_D^1|_{x_*}$. Choose finitely many actual inputs whose values contain a basis, and then apply the stabilizer assertion of Lemma 4.4.

The inputs ϕ_k in this limit need not have a common congruence stabilizer as k varies. The limit takes place in one fiber over D ; a common level is chosen only after finitely many actual inputs have been selected. $\square$

On a compact Kähler quotient a holomorphic one-form is closed and harmonic. A nonzero such form therefore has nonzero cohomology class. These are complex classes. Nothing in this proposition identifies a rational CM source for them.

4.6 Rank-two tensors, wedges, and fixed-data variation

Take $W = W_1 \perp W_2$, so $n = 2$. Project $T_{2,\phi}$ to $\text{Sym}^2((\mathcal{F}_W)_{c\mu}) = \text{Sym}^2(Q^* \otimes W_{\mu}^*)$, and then apply alt_2 . Its target is the canonical bundle in (4.8), up to a constant line. After trivializing that line, denote the resulting differential two-form by $\Xi_W(\phi)$. At a rational decomposition the product formula for theta series will express it as a finite sum of wedges of the one-forms just constructed. We then vary the compact-place data of the total space to compare two such expressions.

We now specify exactly what may vary in this construction. Keep fixed the rational Hermitian space W , the symplectic space $(\mathbb{V}_W, \psi_W)$, the rational embedding $G \rightarrow \text{Sp}(\mathbb{V}_W)$, reference rational Schrödinger coordinates, the global Weil and unitary splitting choices, and the resulting finite action $\omega_{W,f}$. Also fix a single total finite input ϕ in these reference coordinates. At a compact place $\tau \neq \mu$, let

$$W_{\tau} = W_{\tau}^+ \perp W_{\tau}^-$$

range over orthogonal decompositions with

$$\dim W_{\tau}^{\pm} = \#\{i \in \{1, 2\} : d_i(\tau) = \pm 1\}.$$

Require the form on W_{τ}^+ to have sign $-\text{sgn}(t_{\tau})$, and the form on W_{τ}^- the opposite sign. Let $\mathcal{T}$ be the product of these parameter spaces. A definite local factor is a point; when the two signs differ, the local factor is the domain of lines of one prescribed sign in a Hermitian plane of signature $(1, 1)$.

For $t \in \mathcal{T}$, the two compact-place pieces of the positive Lagrangian are

$$V_{\tau} \otimes W_{\tau}^-, \quad V_{\tau}^* \otimes (W_{\tau}^+)^*.$$

In the second expression $(W_{\tau}^+)^*$ is embedded as $\text{ann}(W_{\tau}^-) \subset W_{\tau}^*$; the orthogonal decomposition identifies it with the dual of W_{τ}^+ . The active pieces remain, exactly,

$$L \otimes W_{\mu}, \quad Q^* \otimes W_{\mu}^*.$$

They depend only on W_μ , not on any decomposition of it. These prescriptions give positive Lagrangians for all $t \in \mathcal{T}$. The same projection as in (4.8), with a local smooth choice of constant-line trivialization, defines a smooth form of type $(2, 0)$, denoted $\Xi_W(\phi; t)$, even when t is not induced by rational lines. We do not claim it is holomorphic for such an intermediate parameter.

Proposition 4.6. *The rank-two construction has the following properties.*

- (i) *At period data induced by a rational orthogonal decomposition $W = W_1 \perp W_2$, the form $\Xi_W(\phi)$ is holomorphic. On a sufficiently deep common principal quotient it is a finite sum, with constant coefficients, of forms $\Theta_1(\phi_{1,r}) \wedge \Theta_2(\phi_{2,r})$.*
- (ii) *There is an input which is a product in adapted rational coordinates and for which $\Xi_W(\phi)$ is nonzero.*
- (iii) *Fix W , the total finite input ϕ , and a torsion-free principal quotient X whose finite level fixes ϕ under $\omega_{W,f}$. With locally smooth constant-line trivializations, the map*

$$t \longmapsto \Xi_W(\phi; t)$$

is continuous in the C^∞ topology of differential forms on this fixed compact quotient. The same finite level works throughout the parameter neighborhood.

Proof. For (i), the rational orthogonal decomposition gives a rational symplectic direct sum $\mathbb{V}_W = \mathbb{V}_{W_1} \perp \mathbb{V}_{W_2}$. Choose rational coordinates adapted to it. By Remark 4.3, the total reference input becomes another finite Schwartz input in the adapted coordinates. Such an input is a finite sum

$$\sum_r \phi_{1,r} \otimes \phi_{2,r}.$$

Indeed, a locally constant compactly supported function on the product of two finite-adelic vector spaces is invariant under a sufficiently small product compact open subgroup. Only finitely many of its cosets meet the support, and each coset is a product coset.

For a product input the completed theta series multiply, since both the sum and its completion split in the adapted coordinates. If $T_j^{(i)}$ denotes the degree- j coefficient for the i -th line, the coefficient of total degree two is

$$T_2^{(1)} T_0^{(2)} + T_1^{(1)} T_1^{(2)} + T_0^{(1)} T_2^{(2)}. \quad (4.17)$$

The coefficient of the mixed product is 1, by the $1/k!$ normalization in (4.10). The two unmixed terms have both multiplicity factors in the same one-dimensional $(W_i)_\mu^*$; they vanish under alt_2 . The mixed term becomes the wedge of the two projected first tensors. With compatible constant-line trivializations it is exactly $\Theta_1(\phi_{1,r}) \wedge \Theta_2(\phi_{2,r})$; other trivializations multiply it by a fixed nonzero scalar. Rational changes of coordinates are already accounted for by (4.12). Moreover, two identifications with the differential bundle used here have a ratio invariant under the transitive infinite special-unitary action on D , so that ratio is constant. This proves the wedge expansion and its holomorphy. We have proved holomorphy only after the double-alternating projection: the whole completed second Taylor tensor need not be holomorphic. The finitely many individual inputs descend on a further common principal level by Lemma 4.4.

For (ii), fix a point of D . Proposition 4.5 provides an input for W_1 with a nonzero cotangent value and an input for W_2 whose value is outside the line it spans. Their wedge is nonzero at that point. The product of these inputs in adapted coordinates has a nonzero double-alternating second tensor by the calculation above. Transporting this product back to the reference coordinates gives a total finite input with $\Xi_W(\phi) \neq 0$.

For (iii), no rational coordinate change is made as t varies. The rational symplectic space, its rational G -action, the finite representation, and the total input all remain fixed. The changed compact-place Lagrangians remain invariant under $SU(3)$, since all changes are in the multiplicity factor. Their determinant characters on $SU(3)$ are trivial. The active pieces have not changed, so their determinant character remains ℓ^4 . The infinite comparison in Lemma 4.4 therefore gives the same half-determinant action L^2 at every t . Consequently the same subgroup fixing ϕ gives equivariance and descent at every parameter. There is no finite translation, Hecke choice, or change of level in this variation.

It remains to check uniformity on the quotient. The constant determinant factors in (4.6) vary smoothly with the compact-place subspaces. On a small parameter neighborhood choose smooth nonzero frames and their square roots, agreeing with the chosen frames at the center. Cover the fixed compact X by finitely many charts with precompact lifts to D . On the closure of each lifted chart times a sufficiently small compact parameter neighborhood, the period map is smooth, its image is a compact subset of the Siegel domain, and every fixed order of its derivatives is bounded. The imaginary parts thus have a common positive lower bound. The rational support of the fixed ϕ continues to lie in one fixed finite union of lattice cosets. The Gaussian estimates following (4.10) now give uniform convergence of every fixed order of derivatives in the chart coordinates and the parameters. Projection to (4.8) and the chosen smooth trivializations preserve these bounds. The unchanged equivariance glues these convergent local expressions on X . This is continuity in the C^∞ topology. Compactness of X is used here; no estimate at a cusp or boundary is needed. $\square$

4.7 Approximation while the total theta input stays fixed

We now apply the last proposition to the sign switch. The approximation occurs in $\mathcal{T}$, before the rational coordinate changes used to factor either endpoint.

Corollary 4.7. *Suppose $d_1 + d_2 = d_3 + d_4$, with $d_i(\mu) = 1$ for all i . Start with $W = W_1 \perp W_2$ as above, let t_0 be its compact-place data, and choose a total input ϕ for which $\xi = \Xi_W(\phi; t_0)$ is a nonzero holomorphic top form on a sufficiently deep compact principal quotient X . There are rational orthogonal decompositions $W = W_3 \perp W_4$ with line signs d_3, d_4 whose total compact-place positive Lagrangians are arbitrarily close to those at t_0 . The total active positive Lagrangian is exactly unchanged.*

For a sufficiently close rational choice t' , the form $\xi' = \Xi_W(\phi; t')$ is holomorphic on the same quotient X , with the same total finite input, and

$$\int_X \xi \wedge \bar{\xi}' \neq 0. \quad (4.18)$$

Both endpoints admit the wedge expansions of Proposition 4.6 on a further common principal cover, and their pairing remains nonzero there.

Proof. At a compact place $\tau \neq \mu$, equality of the two sign sums says exactly that the multisets $\{d_1(\tau), d_2(\tau)\}$ and $\{d_3(\tau), d_4(\tau)\}$ coincide. Reorder the two original local lines so that the first has the sign required by $d_3(\tau)$, and choose a nonzero vector on it. After fixing an E -basis of W , weak approximation in E , coordinate by coordinate at these finitely many complex places, gives a single $w \in W(E)$ arbitrarily close to all the chosen local vectors. Nondegeneracy and the signs of their Hermitian norms are open conditions. For a sufficiently close approximation,

$$W_3 = Ew, \quad W_4 = W_3^\perp$$

is therefore a rational orthogonal decomposition. Its first line has the prescribed compact-place signs for d_3 ; orthogonal inertia gives the complementary signs for d_4 . At μ the form on W_μ is definite, so both new lines automatically have the same required sign there.

The total compact-place Lagrangians converge to the original ones. Reordering the original lines only relabels their total positive summands. For a nondegenerate local vector the orthogonal projection onto its line is

$$z \longmapsto h_W(z, w)h_W(w, w)^{-1}w.$$

It varies continuously with w , so both the line and its orthogonal complement converge to the chosen original ones. The selected signed subspaces, and hence the total positive Lagrangians, converge. At the active pair the total pieces are always $L \otimes W_\mu$ and $Q^* \otimes W_\mu^*$, so they are unchanged, not merely close. The approximating rational lines may have small active norms or large arithmetic denominators. No bound on these is needed: the active total period data are fixed, and the continuity assertion concerns the total tensor in the original reference coordinates.

All the hypotheses of Proposition 4.6(iii) now hold with the same rational W , the same G , the same total finite ϕ , and the same compact X . Thus

$$\Xi_W(\phi; t') \longrightarrow \Xi_W(\phi; t_0)$$

smoothly on X . The limit ξ is a nonzero holomorphic top form. In local complex coordinates $\xi = f dz_1 \wedge dz_2$, the form $\xi \wedge \bar{\xi}$ is a positive multiple of $|f|^2 dx_1 \wedge dy_1 \wedge dx_2 \wedge dy_2$. Consequently $\int_X \xi \wedge \bar{\xi} > 0$. Continuity of the integral gives (4.18) for all sufficiently close rational choices. At each such endpoint holomorphy follows from Proposition 4.6(i), not from a holomorphy assertion along the intermediate parameter path.

Factor the two endpoints only now. There are finitely many line inputs in their two wedge expansions. Choose a principal subgroup contained both in the level of X and in their finite stabilizers. It gives a finite unramified cover $\pi : X' \rightarrow X$ on which all individual factors descend. The covering formula is

$$\int_{X'} \pi^* \xi \wedge \overline{\pi^* \xi'} = \deg(\pi) \int_X \xi \wedge \bar{\xi'} \neq 0. \quad (4.19)$$

Both quotients are compact and torsion-free, so there is no cusp term or ramification correction. This proves the last assertion. $\square$

Expansion of the two pulled-back endpoint forms on X' in (4.19) gives (4.1); in particular at least one pair of individual wedge terms has nonzero mixed period. This is the analytic non-vanishing needed for Proposition 3.1. To apply that criterion we must still express the relevant one-form classes as complex linear combinations of images of the selected CM lines under rational Hodge maps, on a common projective principal cover. That arithmetic step is separate from the approximation just proved.

5 Projective ball quotients and a Frobenius identity

The theta construction has supplied complex one-form classes on compact ball quotients. To use the surface criterion, we must identify their rational sources: for the sign function v , the relevant class must lie in the complex span of the images of $U(v)_1$ under rational Hodge maps from $U(v)$. This section and Section 6 prove the following extraction theorem. We first recall its input data from the theta construction.

Retain the notation of Section 4. Thus $E \subset \bar{\mathbb{Q}}$ is Galois CM, $F = E^+$, $G_E = \text{Gal}(E/\mathbb{Q})$, and $c \in G_E$ is complex conjugation. We assume $s = [F : \mathbb{Q}] \geq 2$. The Hermitian E/F -space V has dimension three, signature $(2, 1)$ at $\mu = 1$, and is positive definite at the other real places. Write

$G = \mathrm{SU}(V)$, let D be its ball of negative lines, and retain the fixed $\mathcal{O}_E$ -lattice $\Lambda \subset V$. Fix an odd function $v : G_E \rightarrow \{1, -1\}$ with $v(1) = 1$, and put

$$d(\tau) = v(\tau^{-1}).$$

For one multiplicity line P with these compact-place signs, write $\Theta(\phi)$ for its theta one-form and abbreviate its finite special-unitary action to $\omega_f = \omega_{P,f}$. We work with a finite collection of inputs at a common principal level. The auxiliary prime will be chosen only after that level is fixed.

Theorem 5.1 (Degree-one CM extraction). *Let $v : G_E \rightarrow \{1, -1\}$ be odd with $v(1) = 1$, and use the multiplicity-line theta construction prescribed by $d(\tau) = v(\tau^{-1})$. Given a finite collection of its finite adelic Schwartz inputs, there is a common sufficiently deep principal ball quotient $S_0 = \Gamma \backslash D$ on which every associated theta one-form class belongs to*

$$\mathrm{span}_{\mathbb{C}} \left\{ f(U(v)_1) : f \in \mathrm{Hom}_{\mathrm{Hdg}}(U(v), H^1(S_0, \mathbb{Q})) \right\}. \quad (5.1)$$

The maps in this formula are rational Hodge maps, and their images are viewed after extension to $\mathbb{C}$. The conclusion persists after passage to a deeper principal level.

The quotient S_0 will be a smooth connected projective surface. Corollary 6.7 gives the version needed for the surface criterion: finitely many families with different sign functions satisfy these span conclusions on one projective principal cover of any prescribed torsion-free principal base quotient. The reference moduli construction depends only on V , so the same geometry serves every multiplicity line.

We begin by realizing the principal quotients as components of a full projective moduli space of polarized abelian varieties with E -action. Keeping all components makes the isogeny correspondences into algebraic Hecke operators on the product of their Albanese varieties. Dense ordinary loci then let us prove an integral Frobenius polynomial for these operators. Section 6 combines that polynomial with the active theta parameter to determine the Hodge types of the generalized Hecke spaces containing the theta classes. Scalar descent then proves the extraction theorem.

5.1 A projective moduli space for the reference three-space

The reference abelian varieties use V , not $V \otimes_E P$. For a negative line $L \in D$, prescribe on the complexified homology

$$\mathcal{F}_\mu = L, \quad \mathcal{F}_c = \mathrm{ann}(L) \subset V_\mu^*,$$

where the Hermitian form identifies the conjugate summand with the complex-linear dual. At each other conjugate pair choose, once and for all, the whole summand on one side and zero on the other. These annihilator prescriptions together form a Lagrangian. An imaginary element $\delta_{\mathrm{ref}} \in E^\times$, whose signs are chosen by weak approximation in F , makes them positive for

$$\psi_{\mathrm{ref}}(x, y) = \mathrm{Tr}_{E/\mathbb{Q}}(\delta_{\mathrm{ref}} h_V(x, y)).$$

Multiplying δ_{ref} by a positive integer makes the form integral on Λ .

Here the homology filtration is the kernel of the Lie quotient. If J is the complex structure on real homology, then $z = u + iJu$ lies in that filtration and

$$\psi_{\mathrm{ref}}(z, \bar{z})/i = -2\psi_{\mathrm{ref}}(u, Ju).$$

Thus $-\psi_{\text{ref}}$ is the Riemann form in the positivity convention of Section 4. The corresponding abelian varieties have dimension $3s$, and their Lie ranks are

$$r_\mu = 2, \quad r_c = 1, \quad \{r_\tau, r_{c\tau}\} = \{0, 3\} \quad (\tau \notin \{1, c\}). \quad (5.2)$$

Fix their polarization type.

Choose N sufficiently divisible to give a fine abelian moduli problem, a torsion-free principal lattice level, and a level fixing the chosen finite theta inputs. We will also require the level to kill all possible unitary determinant roots of unity. A marking means a full ordered basis of N -torsion; we allow every pairing matrix for that basis. In particular, transporting it through an isogeny of degree prime to N is allowed even when the isogeny rescales the polarization pairing.

We retain all components so that the full isogeny correspondences can be defined without assuming that the reference component is preserved. Over a number field $K_0 \subset \overline{\mathbb{Q}}$ containing E , consider the reduced scheme S of tuples

$$(B, \lambda, \iota_E, \eta_N).$$

Here B is an abelian variety of dimension $3s$, λ is a polarization of the fixed type, $\iota_E : \mathcal{O}_E \rightarrow \text{End}(B)$ is a unital action whose Rosati adjoint is c , η_N is the full marking, and each Lie summand has the particular rank fixed above (with values displayed in (5.2)). We retain all connected components of this tuple space.

Lemma 5.2. *The scheme S is of finite type, smooth, and projective over K_0 . Its complex connected components are compact ball quotients. For sufficiently divisible N , the component arising from Λ is the quotient of D by its principal special-unitary level.*

Proof. The Lie-rank condition need not identify a single rational Hermitian isomorphism class. We therefore construct the whole finite-type action locus, identify its complex components as ball quotients, and then prove projectivity and identify the component of Λ .

The finite-type action locus. The fine-moduli theorem for polarized abelian varieties gives a quasi-projective scheme with universal polarization and full level at sufficiently large invertible level [31, Chapter 7, Definitions 7.1–7.2, Proposition 7.6, and Theorem 7.9]. In that theorem the polarization degree is fixed and the level is an ordered torsion basis, with no prescribed pairing matrix. After inverting the polarization degree, the polarization kernel is finite étale, so its geometric elementary-divisor type is locally constant. We may therefore take the open-and-closed part of the required type. If necessary, first apply the theorem to a sufficiently large divisor of N prime to the polarization degree and then add the remaining full N -basis as a finite étale level cover. These constructions work over a ring of integers with finitely many primes inverted, as well as in characteristic zero.

We next impose the action by a finite-type construction. Choose finitely many ring generators b of $\mathcal{O}_E$ and parameterize their graphs in relative Hilbert schemes of the square of the universal abelian scheme. The degrees of the graphs in question are bounded. Indeed, put $\alpha = 1 + \bar{b}b$. It is a nonzero totally positive element of F . Choose a positive integer m with $m\alpha^{-1} \in \mathcal{O}_E$. In every such action α is an isogeny, and

$$\deg(\alpha) \deg(m\alpha^{-1}) = m^{6s}.$$

The second factor is a positive integer, so $\deg(\alpha)$ is bounded independently of the point.

To translate this into a Hilbert bound, the normalized Poincaré bundle on the universal abelian scheme times its dual, pulled back by (id, λ) , is a relatively ample bundle inducing 2λ . The tensor product of its two pullbacks to the square restricts on the graph of b to a bundle inducing $2\lambda(1 + \bar{b}b)$.

Its polarization degree is bounded by the preceding identity. Riemann–Roch on an abelian variety of dimension $3s$ gives the Hilbert polynomial $\chi(\mathcal{L})n^{3s}$, with $\chi(\mathcal{L})^2 = \deg(\phi_{\mathcal{L}})$. Only finitely many polynomials can therefore occur. On their graph loci impose the homomorphism, ring, Rosati, and Lie-rank identities. This gives a finite-type action locus.

Over the characteristic-zero polarized base this locus is unramified. One form of rigidity suffices: an infinitesimal difference of two homomorphisms between fixed abelian schemes would give a homomorphism from a proper connected abelian fiber to an additive tangent group, and must vanish. The locus is also proper over that base. Given a discrete valuation ring mapping to the polarized base and a lift of its generic point to the action locus, each generator homomorphism extends uniquely between the fixed abelian schemes by the Néron mapping property [8, §1.2, Definition 1 and Proposition 8]. Its graph remains a graph of the same Hilbert polynomial. The ring and Rosati identities extend by uniqueness. Since the base contains E , the action algebra splits into idempotents in characteristic zero, and their Lie summands have locally constant ranks. Thus the extended point lies in the same action locus. The valuative criterion gives properness. A proper finite-type unramified morphism is quasi-finite and hence finite, so the action locus is finite over the polarized base.

The complex components. We now identify the reduced complex structure of this algebraic locus. In a marked period chart, the integral matrices of the action are locally constant. Such a matrix acts on the complex torus exactly when it preserves the homology filtration; the resulting holomorphic endomorphism of a polarized complex torus is algebraic. Hence the reduced local locus is precisely the locus of positive filtrations stable under the fixed matrices. Trace duality and the Rosati identity express the alternating form as an imaginary trace multiple of an E/F -Hermitian form on the rational homology, of E -dimension three. Isotropy makes the two filtrations in a conjugate pair annihilators. Positivity and (5.2) leave one negative line at the active place and constant data at every other place. The local locus is therefore a smooth complex two-ball.

This description gives whole ball quotients, not only local ball charts. Fix a marked polarized E -lattice occurring in S . Every positive filtration in its compatible ball gives a complex torus with the fixed integral Riemann form, and hence an abelian variety; the preserving integral matrices give its algebraic $\mathcal{O}_E$ -action. Conversely every tuple with these lattice data gives such a filtration. Its connected component is thus the quotient of the whole ball by the marked lattice stabilizer.

Projectivity and the principal level. For each such rational Hermitian space, a vector isotropic over E would remain isotropic at a definite real place. Since $s \geq 2$, there is no such vector, and its special-unitary group is F -anisotropic. The arithmetic lattice and compactness theorems give compact ball quotients [6]; see [27, Theorem 3.3 and Proposition 3.5]. The level removes stabilizers. There are finitely many components because S is of finite type. The finite action locus is quasi-projective, and its reduced complex space is smooth with compact components. It is therefore proper, and a proper quasi-projective scheme is projective. Smoothness and properness descend from the complex base change to K_0 .

Finally the initial lattice Λ is among these data. Even if it is not free, its determinant module is a fractional $\mathcal{O}_E$ -ideal. The determinant of a unitary lattice automorphism is consequently a unit of $\mathcal{O}_E$ of norm one. All its complex conjugates have modulus one, so it is a root of unity. A sufficiently divisible principal level kills every nontrivial such determinant. Its stabilizer is then precisely the principal special-unitary lattice level, as claimed. $\square$

5.2 Good models and ordinary points on every component

Enlarge K_0 to a number field $K \subset \overline{\mathbb{Q}}$ over which all connected components S_j are geometrically connected and have K -points. Index the original ball component by S_0 . Use those points as origins

for the Albanese maps

$$a_j : S_j \longrightarrow A_j = \text{Alb}(S_j), \quad A_S = \prod_j A_j.$$

We regard each A_j as a factor of A_S , with zero in the other factors. The Albanese pullbacks give an isomorphism of rational Hodge structures

$$H^1(A_S, \mathbb{Q}) \simeq H^1(S, \mathbb{Q}). \quad (5.3)$$

Because the Frobenius identity will be checked on ordinary points and then on Albanese differences, we need density on every component and generation of each reduced Albanese factor by those differences.

Lemma 5.3. *After one finite extension of K and removal of finitely many primes, the S_j have smooth proper models with geometrically connected fibers, the A_j extend to abelian schemes, and the a_j extend to the models. The model of S lies in the full tuple parameter scheme, retaining the $\mathcal{O}_E$ -action and its graph coordinates. Differences of Albanese images generate every geometric fiber of each A_j .*

At every remaining rational prime q completely split in K , the ordinary locus of the universal abelian scheme is dense in each geometric component of the special fiber of S .

Proof. Models and Albanese generation. Spread the finitely many schemes, morphisms, and universal data over an open part of the ring of integers of K . One can obtain the proper model of S by taking its closure in a projective compactification of the full parameter scheme used in the proof of Lemma 5.2. Its intersection with the boundary has proper closed image on the integer base. That image misses the generic point because S is proper and lies in the parameter scheme. Removing the finitely many primes in the image leaves the closure inside the full parameter scheme. Thus its geometric points are still distinguished by their polarized, acted-on, marked tuples, including the graph coordinates. Further shrinking gives smooth proper models with geometrically connected component fibers, good abelian models, and the extended Albanese maps.

For a component S_j , let $\Delta_j = a_j(S_j) - a_j(S_j) \subset A_j$. It is closed, irreducible, symmetric, and contains zero. The successive sum images $n\Delta_j$ are nested irreducible closed sets. Their dimensions eventually stabilize; then equality of two successive sets implies stability under addition by Δ_j . Symmetry shows that the stable image is the abelian subvariety generated by Δ_j , which is A_j by the Albanese property. Thus for some fixed n the proper sum map

$$(S_j \times S_j)^n \longrightarrow A_j$$

is surjective. Spread this map together with the other data. Its closed image contains the generic fiber of the flat abelian scheme; that fiber is dense, so the image is the whole abelian scheme. Consequently the sum map is surjective on every geometric fiber.

Ordinary points. For each characteristic-zero component, we now choose a point whose reduction at the split primes retained below is ordinary. In its rational Hermitian three-space choose, by weak approximation, an E -line negative at μ , and diagonalize its orthogonal complement. At the period whose negative line is this coordinate line, the three rational orthogonal projectors split the rational homology into three E -lines and preserve the filtration. Integral multiples of them are endomorphisms of the corresponding abelian variety.

These points and endomorphisms are defined over $\overline{\mathbb{Q}}$. Indeed, add their graphs as extra parameters with the Hilbert polynomials at the selected point. This is a finite-type parameter scheme over a number field. Locally the projector matrices are fixed by rigidity. A negative line preserved by all three projectors must be the unique negative coordinate line, so the selected

enhanced point is isolated. An isolated complex point of a finite-type scheme over a number field has algebraic coordinates. Adjoin the finitely many fields of definition to K , spread these points and endomorphisms, and exclude their projector and isogeny denominators.

Now let q be a remaining completely split prime. At each selected point the integral projectors and the idempotents of $\mathcal{O}_E \otimes \mathbb{Z}_q$ split the q -divisible group into pieces of height one. Their heights are the corresponding characteristic-zero Tate ranks and stay constant under specialization. A height-one q -divisible group over an algebraically closed field has dimension zero or one. In the first case it is étale; in the second its dual has dimension zero, so it is multiplicative. Thus each selected reduction is ordinary. Ordinarity is open, and a smooth geometrically connected fiber is geometrically irreducible. This nonempty open is dense on each component. Chebotarev's theorem supplies infinitely many completely split rational primes after the finite exclusions. $\square$

Choose such a rational prime q , also prime to the polarization degree, N , the discriminant of E , and all exceptional local data of the finite theta inputs. Fix an embedding

$$\iota : \overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}}_q$$

and normalize ν_i by $\nu_i(q) = 1$. The resulting completion of K is $\mathbb{Q}_q$, with residue field $\mathbb{F}_q$. For $\tau \in G_E$, let $\mathfrak{p}_\tau$ be the prime of E defined by $\iota \circ \tau$. Thus for $x \in E^\times$,

$$\nu_i(\tau(x)) = \text{val}_{\mathfrak{p}_\tau}(x).$$

Select one prime from each conjugate pair, including

$$w = \mathfrak{p}_c, \quad \bar{w} = \mathfrak{p}_1. \quad (5.4)$$

Write $\mathcal{J}$ for this set of selected blocks. The w -block has height three and Lie rank one; every other selected block has height three and Lie rank zero or three. The opposite blocks are their polarization duals. These rank assertions persist on the model because the corresponding idempotent Lie and torsion summands are locally free. Choosing $w = \mathfrak{p}_c$, rather than $\mathfrak{p}_1$, will fix the conjugate-valuation signs.

5.3 Algebraic Hecke operators on the Albanese

Choose bases for the selected q -adic lattices and the opposite dual lattices. Since the polarization is prime to q , the split local similitude group and its integral subgroup are

$$\mathcal{G}_q = \left(\prod_{j \in \mathcal{J}} \text{GL}_3(\mathbb{Q}_q) \right) \times \mathbb{Q}_q^\times, \quad \mathcal{K}_q = \left(\prod_{j \in \mathcal{J}} \text{GL}_3(\mathbb{Z}_q) \right) \times \mathbb{Z}_q^\times.$$

The last coordinate is the common polarization multiplier. Use the raw spherical Hecke algebra with $\text{vol}(\mathcal{K}_q) = 1$. Its integral basis consists of double-coset characteristic functions; its rational extension will act on the Albanese up to isogeny. A double coset is specified by a multiplier exponent $h \in \mathbb{Z}$ and, on each selected block, an unordered triple of elementary exponents $(a_{j,1}, a_{j,2}, a_{j,3})$. For a representative g_j with those exponents, our target position is the overlattice $g_j^{-1} \Lambda_j$; branches are represented modulo the integral subgroup on the left.

First suppose that

$$h \geq 0, \quad 0 \leq a_{j,k} \leq h. \quad (5.5)$$

On the j -block of $B[q^h]$, choose a subgroup of the indicated elementary divisors. On the opposite block choose its annihilator under the perfect q^h -torsion pairing. The total subgroup H is maximal

isotropic: each conjugate pair contributes order q^{3h} , so its total order is q^{3sh} . Its quotient $f : B \rightarrow B' = B/H$ is required to have polarization

$$f^\vee \lambda' f = q^h \lambda. \quad (5.6)$$

We verify that this is an algebraic construction in families, including the polarization homomorphism.

Over the characteristic-zero parameter scheme the torsion and the subgroups are finite étale. The translation quotient exists as an fppf scheme quotient for the free finite-locally-free action [36, Tags 07S6 and 07S7]. Its group law descends, and the finite étale quotient of the abelian scheme is again an abelian scheme.

On a complex fiber the polarization in (5.6) is seen directly on lattices. If $\Lambda_j \subset \Lambda'_j \subset q^{-h} \Lambda_j$ is the selected target overlattice and the multiplier is q^h , its opposite lattice is

$$q^{-h}(\Lambda'_j)^\vee.$$

The q^h -scaled alternating form is integral and perfect on this pair at q , by the annihilator condition. At every other prime the q -isogeny changes no elementary-divisor valuation. The scaled form is positive, so it gives a Riemann polarization on the quotient of the original type. More generally, the field-level isotropic descent criterion gives the unique symmetric polarization homomorphism on every geometric fiber [11, Chapter XI, Proposition 11.25].

Here is the relative descent from this fiber calculation. Since $H \subset B[q^h]$, multiplication by q^h descends to $g : B' \rightarrow B$, with

$$gf = [q^h]_B, \quad fg = [q^h]_{B'}.$$

Put $\beta = g^\vee \lambda g : B' \rightarrow B'^\vee$. On each geometric fiber the preceding polarization satisfies $\beta = q^h \lambda'$, so β kills $B'[q^h]$. The correspondence parameter scheme is smooth and reduced, and $B'[q^h]$ is finite étale over it, hence reduced. Vanishing on its geometric points is therefore vanishing of the relative morphism. The fppf quotient by $B'[q^h]$ now factors β uniquely through $[q^h]_{B'}$. It gives a relative homomorphism λ' with $\beta = q^h \lambda'$. Its fibers are the symmetric polarizations just constructed, and cancellation of isogenies gives (5.6). This also proves uniqueness. We have descended the polarization homomorphism, not a particular line bundle.

The $\mathcal{O}_E$ -action descends because H is stable. Cancelling f in (5.6) shows that its Rosati adjoint remains c . The characteristic-zero Lie ranks are unchanged, and f transports the full N -basis. Thus the quotient is a point of the same full tuple space. The subgroup choices form a finite étale scheme over S ; the quotient construction is its algebraic target map. Distinct kernels remain distinct branches even when their target tuples are isomorphic.

Lemma 5.4. *The rational spherical Hecke algebra of $(\mathcal{G}_q, \mathcal{H}_q)$ acts in $\text{End}_K(A_S) \otimes \mathbb{Q}$ and, by pullback, on $H^1(S, \mathbb{Q})$. A basis operator in the positive range (5.5) is an integral homomorphism defined over K . Its pullback on one-forms is the sum of pullbacks over its target branches.*

Proof. Let $C \rightarrow S$ be one of the finite étale source correspondences. On a source component, compose each target branch with the appropriate Albanese map and sum in A_S . This sum descends from a cover trivializing the branches, because it is invariant under their permutations. It is a morphism $n_j : S_j \rightarrow A_S$. Subtracting n_j at the basepoint and applying the Albanese universal property gives an integral homomorphism $A_j \rightarrow A_S$. On differences it has the exact formula

$$T(a_j(x) - a_j(y)) = \sum_{z \in C_x} a_{j(z)}(t(z)) - \sum_{z \in C_y} a_{j(z)}(t(z)), \quad (5.7)$$

where t is the target map, $j(z)$ indexes the component containing $t(z)$, fibers are counted with their branches, and $a_{j(z)}$ is included in A_S . These homomorphisms on the factors define the operator on A_S . Their pullback on one-forms is the branch sum, since a constant translation contributes no differential.

Composition has the raw spherical convolution multiplicities. For a selected lattice Λ_j , a branch is an overlattice $g_j^{-1}\Lambda_j$ with a representative modulo the integral subgroup on the left. If its cumulative multiplier is q^h , the opposite target lattice is forced to be $q^{-h}(\Lambda'_j)^\vee$. Consequently a chain of quotient choices is precisely a chain of selected overlattices, as in spherical convolution. The opposite lattices, polarizations, and transported markings agree for the same composite. The numbers of branches landing in each target component are locally constant on a connected source. Thus the component basepoint translations cancel when (5.7) is composed.

Consider the central position with all selected exponents equal to one and multiplier exponent two. It is quotient by $B[q]$; under the multiplication-by- q identification its target has the same polarization and action and marking $q\eta_N$. It is therefore a finite-order marking automorphism, since q is invertible modulo N . Adding a sufficiently large power of this position to any integer exponent position puts it in (5.5). Define its operator by clearing that power with the inverse marking automorphism. The positive convolution identity shows that this definition is independent of the clearing power. This gives the full rational spherical action. The spherical algebra is commutative, as follows also from the Satake isomorphism [14, §3]. $\square$

5.4 A Frobenius polynomial from ordinary neighbors

The Hecke action is now algebraic. We next relate its reduction to Frobenius by counting quotient isogenies at ordinary points of the universal family. This calculation depends on the reference three-space and its Lie ranks; it is independent of the theta multiplicity line.

For $i = 1, 2, 3$, let T_i be the full moduli Hecke operator with i active exponents equal to one, all other selected exponents zero, and multiplier exponent zero. These are the active minuscule cosets in the full similitude algebra. They need not be positive in (5.5); Lemma 5.4 defines them by central clearing.

Let C_0 be the central position with multiplier exponent one and all active exponents zero. On a nonactive selected block, take all three exponents to be one if its Lie rank is three, and zero if its Lie rank is zero. Put

$$D_i = C_0^i T_i, \quad 1 \leq i \leq 3. \quad (5.8)$$

Here D_i has multiplier exponent i , active exponents zero or one, and nonactive selected exponents zero or i . It is a positive raw operator, hence an integral homomorphism of A_S . The quotient branches defining D_i can therefore be studied by specializing the isogenies constructed in Subsection 5.3.

Let $\tilde{A}_S/\mathbb{F}_q$ be the good reduction at the fixed completion of K , and let Φ be its q -power Frobenius endomorphism. The K -defined integral raw Hecke homomorphisms extend over $\mathbb{Z}_q$ by the Néron property; rational combinations extend after tensoring the endomorphism ring with $\mathbb{Q}$. Their reductions are defined over $\mathbb{F}_q$, so they commute with Φ . Use the same letters D_i for these reductions.

Proposition 5.5. *In the integral endomorphism ring of $\tilde{A}_S$,*

$$\Phi^3 - D_1\Phi^2 + qD_2\Phi - q^3D_3 = 0. \quad (5.9)$$

Proof. Let $k = \overline{\mathbb{F}_q}$, and let x be an ordinary geometric point of the special fiber of S . Its active q -divisible group has the canonical splitting

$$G_w = M_0 \oplus E_0,$$

where M_0 is multiplicative of height one and E_0 is étale of height two. Over the perfect field k , the reduced subgroup at each finite torsion level is the canonical étale part; it is preserved by the endomorphism idempotents. Each nonactive selected block is wholly multiplicative or wholly étale according to its Lie rank.

Three quotient constructions will account for the neighbors. Write X_0 for the quotient with active kernel $M_0[q]$ and multiplier q , Y_0 for the sum of quotients with active kernel a line $\ell \subset E_0[q]$ and multiplier q , and Z_0 for the quotient with active kernel $E_0[q]$ and multiplier q^2 . For multiplier q^h , prescribe the connected q^h -torsion on every nonactive selected block and the forced annihilator on every opposite block. We will prove that these quotients define full tuples of S , that X_0 is Frobenius, and that the specialized neighbor sums are

$$D_1 : X_0 + qY_0, \quad D_2 : X_0Y_0 + q^2Z_0, \quad D_3 : X_0Z_0.$$

The displayed formulas are the identities to establish; at this point the quotient constructions have not yet been identified with operations on S . Once they are, their cancellation will give the asserted polynomial on Albanese differences.

Specialization multiplicities. Lift x , using smoothness, to $R = W(k)$, and write $\mathcal{G}_w$ for the lifted active q -divisible group. The ring R is a complete henselian discrete valuation ring. In the lifted active q -torsion, the q^2 open-and-closed special label components lift uniquely by lifting idempotents of its finite coordinate algebra. Each lifted component is finite flat of rank q . The identity-specializing component is a subgroup, and multiplication respects the lifted labels by uniqueness. Thus there is an exact sequence

$$0 \longrightarrow \mathcal{H}^0 \longrightarrow \mathcal{G}_w[q] \longrightarrow (\mathbb{F}_q^2)_R \longrightarrow 0$$

with $\mathcal{H}^0$ finite flat of rank q . No splitting of this sequence over R is asserted.

Pass to a finite extension of R over which the generic torsion is split. Its active fiber is a three-dimensional $\mathbb{F}_q$ -space with distinguished line $\mathcal{H}_{\text{gen}}^0$. For a fixed b -dimensional subspace of the label quotient, the generic subspaces disjoint from that line and mapping onto the label subspace are its graph lifts. They form a torsor under the b -dimensional space of maps to the line, so there are q^b of them. A generic subspace containing the line is the unique full inverse image of its label subspace.

The schematic closure of a generic subgroup in the finite flat torsion scheme is finite torsion-free over the new DVR, hence flat. It is still a subgroup, since the subgroup identities extend from the generic fiber on flat schemes. Intersection with a label component is an idempotent direct summand of its coordinate algebra, so its rank is its generic rank. A graph has rank one on each of its labels. A finite flat rank-one algebra over a local ring is that ring, and its special fiber is the unique reduced point of that label component. All graph lifts over a label subspace therefore specialize to the same étale subgroup of $E_0[q]$. A subspace containing the distinguished line fills the corresponding rank- q components and specializes to $M_0[q]$ plus that étale subgroup.

The active kernels of D_i are i -dimensional subspaces of the generic q -torsion. The preceding count gives:

operator	special active kernel	multiplicity
D_1	$M_0[q]$	1
	$\ell \subset E_0[q], \dim_{\mathbb{F}_q} \ell = 1$	q for each ℓ
D_2	$M_0[q] \oplus \ell, \dim_{\mathbb{F}_q} \ell = 1$	1 for each ℓ
	$E_0[q]$	q^2
D_3	$M_0[q] \oplus E_0[q]$	1.

In the first two cases the total number is $q^2 + q + 1$: there are $q + 1$ étale lines. The nonactive selected kernels are uniquely their connected q^i -torsion for D_i .

Full tuples and Frobenius. We now show that the counted kernels give targets in S , and identify the resulting quotient operations. Properness of the model of S extends each generic neighbor over a finite valuation-ring extension. The universal source and target are abelian schemes over that new DVR. Apply the Néron mapping property there, anew, to extend the generic isogeny [8, §1.2, Proposition 8]. The polarization identity extends by separatedness. On each fiber the kernel of the extension is contained in the finite kernel of $f^\vee \lambda' f = [q^i] \lambda$, so it has finite kernel and is an isogeny between equal-dimensional abelian varieties. Since source and target polarizations have the same degree, its degree is q^{3si} on every fiber. Properness and fiberwise quasi-finiteness make the extension finite. Fiberwise flatness of isogenies between the smooth fibers and the relative flatness criterion make it flat; see [8, §7.3, Lemma 1 and Proposition 6; §2.4, Proposition 2]. Its finite flat kernel is therefore the schematic closure of its generic kernel.

Finite-locally-free Cartier duality commutes with base change. For a subgroup $H \subset G$ in a finite group with a perfect pairing, the annihilator is

$$\text{ann}(H) = \ker(G^D \longrightarrow H^D) = (G/H)^D.$$

It is finite locally free and commutes with the above specialization. Thus the opposite generic kernels specialize to the annihilators of the selected kernels just counted. The corresponding nonactive kernels are already prescribed. Equal total kernels give canonically isomorphic abelian quotients. Their descended $\mathcal{O}_E$ -actions and transported markings are unique, and their polarizations are unique by cancelling f in $f^\vee \lambda' f = q^i \lambda$. The model retains these full tuples, so equal quotients are equal target points of S . In particular every counted kernel has such a target. It is ordinary, since ordinarity is invariant under isogeny.

The constructions X_0, Y_0, Z_0 are therefore operations on the ordinary points of the same full moduli space. We use them as multiset operations on points; no global special-fiber algebraic correspondence is required.

The operation X_0 is the q -power Frobenius on full tuples. Indeed its total kernel is the connected q -torsion on every block. On the active pair, duality pairs the multiplicative part with the opposite étale part and the étale part with the opposite multiplicative part, so the annihilator of $M_0[q]$ is exactly the opposite connected part; the pure nonactive pairs have the same property. The total kernel is therefore the kernel of relative Frobenius $F_B : B \rightarrow B^{(q)}$. Naturality of the polarization and Frobenius–Verschiebung duality give

$$F_B^\vee \lambda^{(q)} F_B = V_{B^\vee} F_{B^\vee} \lambda = [q] \lambda.$$

Here $V_{B^\vee}$ denotes Verschiebung. The action on the target is $\iota_E^{(q)}$. For the constant label group $T = (\mathbb{Z}/N\mathbb{Z})^{6s}$, naturality gives

$$F_B \eta_N = \eta_N^{(q)} F_T.$$

Relative Frobenius on T fixes its labels, and F_B is an isomorphism on N -torsion. Thus the transported marking is exactly the twisted marking, with no multiplication-by- q marking change. Since the model and universal data are defined over $\mathbb{F}_q$, this is its coordinate- q -power operation on geometric points, not its inverse.

Frobenius commutes with Y_0 and with Z_0 as multiset operations. For any of their quotient maps $f : B \rightarrow B'$,

$$F_{B'} f = f^{(q)} F_B.$$

Frobenius is an isomorphism on the étale part and permutes its line choices bijectively; this square also respects the polarizations, actions, and markings. On the active block the composite X_0Y_0 has kernel $M_0[q] \oplus \ell$ with multiplier q^2 , and X_0Z_0 has kernel $M_0[q] \oplus E_0[q]$ with multiplier q^3 .

For clarity, the opposite kernels of these composites are indeed forced by their selected kernels and multipliers. On the q -divisible groups, write G, G' for source and target and define the polarization adjoint $f^\dagger = \lambda^{-1} f^D \lambda'$; the polarizations are isomorphisms there because their degrees are prime to q . For a composite with multiplier q^h ,

$$f^\dagger f = [q^h], \quad f f^\dagger = [q^h].$$

The second identity follows by cancellation of the fppf epimorphism f . As fppf subgroup sheaves, $\ker f = f^\dagger(G'[q^h])$. For $u, v \in G'[q^h]$, adjunction gives

$$e_{q^h}(f^\dagger u, f^\dagger v) = e'_{q^h}(u, f f^\dagger v) = 1,$$

so the image is isotropic in $G[q^h]$. The endomorphism idempotents split its kernel into conjugate-block parts. If $K_j, K_{\bar{j}}$ are the parts in one conjugate pair of height-three blocks, then

$$K_{\bar{j}} \subseteq K_j^\perp, \quad \text{rank}(K_j) \text{rank}(K_{\bar{j}}) \leq q^{3h}.$$

Cartier duality gives equal kernel ranks for f and $f^\dagger$. Their composite has kernel rank q^{6sh} , so $\text{rank ker } f = q^{3sh}$. Multiplying the pairwise inequalities gives equality, and hence equality in each pair. Thus $K_{\bar{j}} = K_j^\perp$. For X_0Z_0 , for example, the annihilator of the active q -torsion in q^3 -torsion is the opposite block's q^2 -torsion. The pure nonactive connected kernels likewise compose to connected q^h -torsion. Uniqueness of the polarizations and markings now identifies the composite targets with the counted targets. We have proved the multiset identities

$$\begin{aligned} D_1 : & X_0 + qY_0, \\ D_2 : & X_0Y_0 + q^2Z_0, \\ D_3 : & X_0Z_0. \end{aligned} \tag{5.10}$$

The integral Albanese identity. The neighbor counts are now identities of multisets of full tuples. We pass from them to endomorphisms by applying the Albanese difference formula. Let $\mathcal{O}$ be the set of ordinary k -points of the union of components and extend X_0, Y_0, Z_0 linearly to the free abelian group $\mathbb{Z}[\mathcal{O}]$. Define

$$\alpha : \mathbb{Z}[\mathcal{O}] \longrightarrow \tilde{A}_S(k), \quad \alpha([x]) = a_{j(x)}(x).$$

The component Albanese maps reduce over $\mathbb{F}_q$, so $\Phi\alpha = \alpha X_0$. For ordinary x, y in one source component, lift them and specialize the integral identity (5.7). With

$$N_1 = X_0 + qY_0, \quad N_2 = X_0Y_0 + q^2Z_0, \quad N_3 = X_0Z_0,$$

it gives

$$D_i(a_j(x) - a_j(y)) = \alpha(N_i([x] - [y])).$$

This is an equality in the actual group of geometric points, with all branch multiplicities retained. No rationalization of a finite-field point group is used.

Frobenius preserves each component and its ordinary locus. Applying the same formula to $X_0^m x, X_0^m y$ shows that the endomorphism in (5.9) on this difference is

$$\alpha((X_0^3 - N_1 X_0^2 + q N_2 X_0 - q^3 N_3)([x] - [y])).$$

The operator inside this expression vanishes:

$$\begin{aligned} X_0^3 - N_1 X_0^2 + q N_2 X_0 - q^3 N_3 \\ = -q Y_0 X_0^2 + q X_0 Y_0 X_0 + q^3 Z_0 X_0 - q^3 X_0 Z_0 = 0. \end{aligned}$$

Only the two proved commutations with X_0 are used. Ordinary points are dense in every smooth component by Lemma 5.3. Therefore the morphism obtained by applying the endomorphism to $a_j(x) - a_j(y)$ vanishes on all of the product of that component with itself. The finite-sum generation in the same lemma makes the endomorphism zero on each Albanese factor. It is zero on $\tilde{A}_S$ after base change to k ; faithful base change gives the integral identity over $\mathbb{F}_q$. $\square$

For the cohomological application, we record the Frobenius convention. On $H_{\text{cris}}^1(\tilde{A}_S/\mathbb{Z}_q)[1/q]$, crystalline Frobenius is the pullback $\varphi = \Phi^*$ of this q -power endomorphism: it is defined by absolute-Frobenius pullback, and the Witt Frobenius on $W(\mathbb{F}_q) = \mathbb{Z}_q$ is the identity because q is prime [36, Tag 07N0]. In degree-one cohomology pullback is additive in abelian homomorphisms. It reverses composition, but the displayed endomorphisms commute. Thus (5.9) gives the same operator polynomial with φ and the Hecke pullbacks, not a reciprocal polynomial. This is the identity used in Proposition 6.6.

6 CM sources of the theta one-forms

We now prove Theorem 5.1, which identifies the rational CM sources of the theta classes. Section 5 constructed an algebraic Hecke action and a cubic relation with Frobenius. We calculate the active Hecke characters of the theta classes, insert them into that relation, and use degree-one Weil weight to isolate one Frobenius slope on every full character space. The Hecke projectors preserve the Hodge filtration, so these slopes determine the Hodge types of all scalar conjugates. Scalar descent will then give the required rational Hodge maps.

Retain the fields, moduli space, Albanese product, and Hecke operators of Section 5. In particular, the split good prime q was chosen after fixing the finite theta inputs and their common principal level.

6.1 The active spherical parameter

We calculate the active Hecke characters in two steps. First we identify the geometric Hecke action on theta inputs as the inverse spherical action. We then compute that action on local inputs fixed by $\text{SL}_3(\mathbb{Z}_q)$, which form a cyclic module for a dilation operator. Rational scalar covariance will make its eigenvalues algebraic and encode the compact-place signs in their q -adic valuations. A lattice count then identifies the spherical parameter, with the raw Hecke normalization fixed.

Let $\mathcal{H}_w^0$ be the rational subalgebra supported on positions with multiplier exponent zero, every nonactive selected block trivial, and active exponent sum zero. It is the spherical algebra of $\text{SL}_3(\mathbb{Q}_q)$.

Lemma 6.1. *Every double coset in $\mathcal{H}_w^0$ has all its branches within each component of S . On the original ball component, its pullback action on theta classes is the inverse spherical action on the active local Weil input: a local double coset acts through its inverse.*

Proof. As a quasi-isogeny, such a position leaves the rational polarized period fixed. It changes only the active lattice and its opposite dual, keeping the marking and all other local data. This is also the description obtained by clearing the central shift in Lemma 5.4 and undoing its marking

change. A determinant-exponent-zero lattice position has a representative in $\mathrm{SL}_3(\mathbb{Q}_q)$. Adjusting determinant units in Smith bases shows that its double cosets and branches are exactly those for $\mathrm{SL}_3(\mathbb{Z}_q) \backslash \mathrm{SL}_3(\mathbb{Q}_q) / \mathrm{SL}_3(\mathbb{Z}_q)$.

For each component apply strong approximation to the special-unitary group of its rational Hermitian space. This group is simply connected and absolutely almost simple over F , and its factor at μ is noncompact. Strong approximation away from all infinite places therefore makes its F -points dense in its finite adelic points [27, Theorem 4.16]. Put $K_q = \mathrm{SL}_3(\mathbb{Z}_q)$. Write $\Lambda^{(j)}$ for the marked lattice of the component under consideration. For a branch with new active lattice $g^{-1}\Lambda_w^{(j)}$, write $(\Lambda^{(j)})'$ for the full changed lattice and choose a rational representative γ whose active component lies in $K_q g$ and whose other finite components are integral and marking-preserving. Then $\gamma(\Lambda^{(j)})' = \Lambda^{(j)}$, and γ identifies the target tuple with the transformed negative period line in the same component.

On the original component, Lemmas 4.2 and 4.4 give

$$\gamma^* \Theta(\omega_f(\gamma)\phi) = \Theta(\phi).$$

Thus fixed-input pullback by the branch acts through $\omega_f(\gamma)^{-1}$. Away from w the chosen level fixes the input. At w our good-prime choice makes it invariant under K_q ; if $\gamma_w = kg$, its inverse action on that input is therefore $\omega_f(g^{-1})$. Summing the branches proves the inverse spherical formula. The sum is raw: no division by its degree and no additional power of q occurs. $\square$

We now compute this inverse spherical action on the local inputs.

At w , choose the Schrödinger position space to be the three-dimensional w -embedding summand, with lattice $\mathbb{Z}_q^3$. By Lemma 4.4, the action of $\mathrm{SL}_3(\mathbb{Q}_q)$ is inverse linear substitution. Every finite input is a finite sum of tensors separating this place from the other finite places. Our choice of q makes the input invariant under $\mathrm{SL}_3(\mathbb{Z}_q)$; averaging a separated decomposition over that compact group gives one whose local factors are invariant.

The invariant Schwartz functions are radial: $\mathrm{SL}_3(\mathbb{Z}_q)$ is transitive on vectors with a fixed minimum coordinate valuation. Local constancy at zero and compact support therefore make this space the span of

$$b_j = \mathbf{1}_{q^j \mathbb{Z}_q^3}, \quad j \in \mathbb{Z}.$$

The functions b_j are linearly independent. Define the normalized dilation

$$(\mathcal{D}_q b)(x) = q^{3/2} b(q^{-1}x). \tag{6.1}$$

Then $\mathcal{D}_q b_j = q^{3/2} b_{j+1}$, so the radial space is a cyclic module for $\mathbb{C}[\mathcal{D}_q, \mathcal{D}_q^{-1}]$.

Lemma 6.2. *Fix the input at all finite places other than w . There are $M > 0$, an element $r \in E^\times$, and a nonzero algebraic number ω such that*

$$r\bar{r} = 1, \quad \mathrm{val}_w(r) = M, \quad \mathrm{val}_{\bar{w}}(r) = -M,$$

with zero valuation at every other finite prime, and the map from radial inputs to theta one-form classes factors through $\mathcal{D}_q^M = \omega$. In the covariance convention of Lemma 4.2,

$$\omega^2 = \zeta \prod_{\tau \in \Sigma} \tau(r)^{-3d(\tau)} \tag{6.2}$$

for a root of unity ζ . Every root of $X^M - \omega$ has absolute value one under every complex embedding.

Proof. Principalize a positive power $w^m = (a)$ and put $r = a/\bar{a}$. It has norm one and the required valuations with $M = m$. At every other finite place it is a unit. Powers of an element of a compact unit group enter every prescribed open subgroup: its image in the finite quotient by that subgroup has finite order. Smoothness of the bare local Weil representation gives an open stabilizer of the fixed input in the metaplectic group; its image meets the scalar unit group in an open subgroup. Replacing r by a common positive power therefore places it in the stabilizers needed at the finitely many exceptional places of the fixed input; replace M accordingly.

Multiplication by r on $V \otimes_E P$ preserves its F -symplectic form because $r\bar{r} = 1$. We use its *bare rational symplectic* covariance, with paired metaplectic lifts, not the value of a chosen unitary splitting character. For a transformation h of a local position space, the normalized linear action is

$$(\rho(h)b)(x) = |\det h|_q^{-1/2} b(h^{-1}x),$$

up to the finite Weil phase of the bare lift. The linear-subgroup formula, rational summation invariance, and the double-cover control are in [37, I, no. 13; III, no. 41; IV, nos. 43–45]. At w , the position action is the scalar r_w , of valuation M . Its unit part fixes a radial function, so the finite action there is $\mathcal{D}_q^M$, up to a root of unity. At almost every other place the input is the unramified lattice vector and r is integral. At the remaining places the chosen power fixes the input under a suitable bare lift; the paired lift differs only by its finite phase. Their product is again a root of unity. An arbitrary lift in the circle pushout would not justify this conclusion, which is why the bare rational action is used here.

At infinity, multiplication by r acts on the τ -embedding summand as the scalar $\tau(r)$: this is the defining scalar action after tensoring the E -space through τ . On the $c\tau$ -summand it acts as $\tau(\bar{r}) = \tau(r)^{-1}$. It consequently fixes every chosen positive Lagrangian and the point of D . In the identity $ge_Z = e_{gZ}J$ of Lemma 4.2, J is this action on the homology filtration. The squared scalar on the projected first tensor is the scalar on $\det \mathcal{F}_P$ times the square of the scalar on its projected c -summand. At μ , the filtration has scalar exponents $1, -1, -1$. Its determinant contributes -1 , and the squared projected conjugate factor contributes -2 . The total is -3 . At a compact pair, the selected full rank-three summand contributes $-3d(\tau)$, with no projected tangent factor. Thus the squared infinite scalar is $\prod_{\tau \in \Sigma} \tau(r)^{-3d(\tau)}$. This calculation includes the scalar action on the determinant's constant line, which is trivial only on $SU(V)$.

The covariance formula has the finite transformed input on its left and the tensor action on its right. Since r fixes the period, it gives $\Theta(\mathcal{D}_q^M b) = \omega \Theta(b)$, with the finite phase absorbed in ω , and its square is exactly (6.2). This equality of one-forms implies the asserted equality of cohomology classes.

The right side of (6.2) is algebraic, so ω is algebraic. Every complex embedding of E sends r to a number of modulus one, since E is CM and $r\bar{r} = 1$. Applying any scalar conjugation to the displayed identity shows that every complex conjugate of ω has modulus one. The same is then true of each root of $X^M - \omega$. $\square$

For a finite separated sum of inputs, take the common stabilizers of all its other-place factors in this proof. The same r, M, ω then works for that sum.

We use the half-modulus Satake convention, namely the parameter attached to normalized parabolic induction. We do not apply the additional central Hecke/Tate shift of Gross's Section 8. In particular the trivial representation of GL_2 has parameter $\{q^{1/2}, q^{-1/2}\}$. On SL_3 , a parameter is an unordered triple of nonzero scalars modulo a common scalar. We say that an active character occurs in a theta class when its simultaneous generalized eigenspace in the finite-dimensional Hecke span has a nonzero projection of that class.

Proposition 6.3. *Every active spherical character occurring in a theta class of the multiplicity line P has half-modulus projective parameter*

$$\{q^{1/2}, q^{-1/2}, e\}, \quad (6.3)$$

where $e = e_0^{-1}$ for a root $e_0^M = \omega$ from Lemma 6.2. It is algebraic and all its complex conjugates have modulus one. For every $\sigma \in \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$,

$$\nu_i(\sigma e) = -\frac{3}{2}d((\sigma|_E)^{-1}) = -\frac{3}{2}v(\sigma|_E). \quad (6.4)$$

A scalar-conjugate active character has a representative of the form (6.3) with third entry $e' = \pm \sigma e$.

Proof. The active spherical action commutes with $\mathcal{D}_q$. The radial Laurent module modulo $\mathcal{D}_q^M - \omega$ is a direct sum of one-dimensional spaces $\mathcal{D}_q = e_0$, because $X^M - \omega$ has distinct nonzero roots. It suffices to compute the spherical character on one of them. This decomposition concerns the radial quotient only; it asserts no semisimplicity for the ambient cohomological Hecke action.

For this computation, extend the substitutions to

$$\rho(g)b(x) = |\det g|_q^{-1/2} b(g^{-1}x), \quad g \in \text{GL}_3(\mathbb{Q}_q),$$

and put $K_3 = \text{GL}_3(\mathbb{Z}_q)$. This is a computational extension of the active SL_3 -action, not a claim of theta equivariance for the full moduli similitude group. Let T_i be the raw double coset of $\text{diag}(q, \dots, q, 1, \dots, 1)$, with i entries q , and $\text{vol}(K_3) = 1$. Its right cosets correspond to sublattices between $q\mathbb{Z}_q^3$ and $\mathbb{Z}_q^3$ of index q^i . For $i = 1, 2$ there are $q^2 + q + 1$ such lattices. The numbers containing a fixed primitive vector are respectively $q + 1$ and 1. Since $\rho(g)b_0$ is $q^{i/2}$ times the characteristic function of its sublattice, we obtain

$$\begin{aligned} T_1 b_0 &= q^{1/2}((q+1)b_0 + q^2 b_1), \\ T_2 b_0 &= q(b_0 + (q^2 + q)b_1), \\ T_3 b_0 &= \mathcal{D}_q b_0. \end{aligned} \quad (6.5)$$

For example, vectors in $q\mathbb{Z}_q^3$ belong to every sublattice, while a primitive vector is tested by its nonzero class in $\mathbb{F}_q^3$; the displayed counts are the numbers of the corresponding planes or lines containing that class.

On the quotient $\mathcal{D}_q = e_0$ we have $b_1 = q^{-3/2}e_0 b_0$. Substitution into (6.5) gives the values

$$q e_1(e_0, q^{1/2}, q^{-1/2}), \quad q e_2(e_0, q^{1/2}, q^{-1/2}), \quad e_3(e_0, q^{1/2}, q^{-1/2}),$$

where e_i is the i -th elementary symmetric polynomial. In the half-modulus Satake isomorphism the raw minuscule cosets have transforms $q e_1, q e_2, e_3$; these generate the spherical algebra, with e_3 invertible. See [14, Equation (3.14) and §6], where the GL_n formula is attributed to Tamagawa. The character just computed therefore has parameter $\{e_0, q^{1/2}, q^{-1/2}\}$.

The geometric action on theta inputs is inverse by Lemma 6.1. Inversion sends the half-modulus parameter to its reciprocal, without a degree factor. One can check this on the same generators: $T_1^\vee = T_3^{-1}T_2$, $T_2^\vee = T_3^{-1}T_1$, and $T_3^\vee = T_3^{-1}$, where $f^\vee(g) = f(g^{-1})$. Their transforms are the corresponding elementary symmetric functions of the reciprocal triple. After interchanging the first two entries, the active projective parameter is therefore $\{q^{1/2}, q^{-1/2}, e_0^{-1}\}$.

The determinant-zero GL_3 and SL_3 double cosets agree as in Lemma 6.1. Their Satake unipotent integrals and modulus factors agree as well. The half-sum of positive roots is $(1, 0, -1)$, integral on the relevant cocharacter lattices; the Satake comparison is consequently defined over rational

coefficients [14, Proposition 3.6]. Scalar conjugation of an active numerical character therefore conjugates its projective parameter. It may change the two entries $q^{1/2}, q^{-1/2}$ by their common sign. Removing that sign projectively leaves third entry $\pm\sigma e$.

It remains to calculate the valuation. Put $g = \sigma|_E$. By the definition of the prime labels,

$$\nu_l(\sigma\tau(r)) = \text{val}_{\mathfrak{p}_{g\tau}}(r).$$

This is nonzero only when $g\tau = 1$ or c . Among the representatives Σ , the contributing one is g^{-1} with valuation $-M$, or cg^{-1} with valuation M . Here c is central in G_E , and oddness of d gives in either case

$$\sum_{\tau \in \Sigma} d(\tau) \nu_l(\sigma\tau(r)) = -Md(g^{-1}).$$

Taking valuations in (6.2) and in $e_0^M = \omega$ yields $\nu_l(\sigma e_0) = \frac{3}{2}d(g^{-1})$. Since $e = e_0^{-1}$ and $d(g^{-1}) = v(g)$, this proves (6.4). The unit-modulus assertion follows from Lemma 6.2. $\square$

The active character is now known, including the valuation of every scalar conjugate. The full Hecke character may still multiply its three Satake entries by a common scalar. The following finite-order relations will control that scalar when we apply the Frobenius polynomial of Proposition 5.5.

Lemma 6.4. *The central operators T_3 and C_0 have finite order on the full tuple space S and on A_S .*

Proof. Each central position has a unique lattice choice. For T_3 , use the norm-one element r constructed in Lemma 6.2; only its valuations and norm are needed here. The scalar r takes the T_3^M -target lattice back to the source lattice on the active and opposite blocks. It preserves every polarized period because it is an E -scalar of norm one. Its denominators are supported at q , so a further power is one on the N -marking. That power of T_3 is the identity on every tuple.

For C_0 , let I be the product of the primes where its full exponent is one. There is exactly one from each conjugate pair, including $\bar{w}$, and $I\bar{I} = (q)$. Choose $I^m = (a)$. Then

$$a\bar{a} = q^m u$$

for a totally positive unit $u \in \mathcal{O}_F^\times$. The scalar $b = a^2/u$ has exactly the valuations of I^{2m} and satisfies

$$b\bar{b} = q^{2m}.$$

It identifies the corresponding powered target lattice with the source, with exactly the required polarization multiplier. A further power fixes the marking. This argument works on every rational Hermitian component because an E -scalar preserves each of its embedding filtrations. Thus a power of C_0 is also the identity on the full tuple space. The induced Albanese homomorphisms have finite order as well. $\square$

6.2 From Frobenius slopes to Hodge types

We now combine the Frobenius polynomial with the theta parameters to determine the Hodge types of the relevant Hecke spaces. The polynomial will force a single Newton slope on each full generalized character space. To turn this into a Hodge type, we also need its projector to preserve the Hodge filtration. We first construct these projectors and explain the filtered argument.

Let $\mathbb{T}$ be the image of the full spherical Hecke algebra in $\text{End}_{\mathbb{Q}} H^1(A_S, \mathbb{Q})$. It is a finite-dimensional commutative $\mathbb{Q}$ -algebra; its active subalgebra is the image of $\mathcal{H}_w^0$. A numerical character is a $\mathbb{Q}$ -algebra homomorphism $\chi : \mathbb{T} \rightarrow \overline{\mathbb{Q}}$. Inside a character value, the symbol of an endomorphism denotes its degree-one pullback. After a finite extension of algebraic coefficients, $\mathbb{T}$ is

a product of local Artinian algebras. Its product idempotents p_χ give the simultaneous generalized character spaces

$$H_\chi = p_\chi H^1(A_S, \overline{\mathbb{Q}}).$$

If η is a character of the active subalgebra, its generalized space is

$$H_\eta^{\text{act}} = \bigoplus_{\chi|_{\text{act}}=\eta} H_\chi. \quad (6.6)$$

Indeed the image of the active idempotent in a local factor of the full algebra is either zero or one, as decided by its residue character. Thus the sum includes every full extension of η and all of its nilpotent parts. No semisimplicity of the Hecke operators is assumed.

Each Hecke operator is induced by a rational endomorphism of A_S defined over K . In degree-one Betti, de Rham, and crystalline cohomology, pullback is additive in abelian homomorphisms: in the Künneth decomposition of $H^1(A \times A)$, pullback by addition has both components equal to the identity, as restriction to the two axes shows. Pullback reverses composition, but the Hecke endomorphisms commute. Pullback on Betti H^1 is faithful by Lemma 2.2, so polynomial relations in $\mathbb{T}$ are also relations among the corresponding endomorphisms. The same degree-one functoriality preserves those relations in de Rham and crystalline cohomology. Thus the p_χ remain idempotents in these degree-one realizations after extension of coefficients. They need not have rational coefficients; what matters is that they are algebraic-coefficient polynomials in extending endomorphisms.

The following lemma applies to precisely these projected summands. Its use of both complementary projectors is what turns weak admissibility into equality of the Hodge and Newton numbers on each summand.

Lemma 6.5. *Let $\mathcal{A}/\mathbb{Z}_q$ be an abelian scheme with generic fiber A , and put*

$$D_A = H_{\text{dR}}^1(A/\mathbb{Q}_q).$$

Equip D_A with its Hodge filtration and its good-reduction crystalline Frobenius φ_A . Then D_A is weakly admissible, its filtration has only indices 0 and 1, and endomorphisms extending over $\mathbb{Z}_q$ preserve the filtration and commute with φ_A .

Let $L/\mathbb{Q}_q$ be finite. Suppose that an L -linear idempotent p on $D_A \otimes_{\mathbb{Q}_q} L$ preserves the filtration and commutes with $\varphi_A \otimes 1$. If every Frobenius slope on its image is $t \in \{0, 1\}$, then the induced filtration satisfies

$$F^1(p(D_A \otimes L)) = \begin{cases} 0, & t = 0, \\ p(D_A \otimes L), & t = 1. \end{cases}$$

Proof. Good-reduction crystalline comparison identifies

$$D_{\text{cris}}\left(H_{\text{ét}}^1(A_{\overline{\mathbb{Q}}_q}, \mathbb{Q}_q)\right) \simeq H_{\text{cris}}^1(\mathcal{A}_{\mathbb{F}_q}/\mathbb{Z}_q)[1/q]$$

and identifies the filtered de Rham realization with D_A . These identifications are functorial for extending endomorphisms and compatible with Frobenius and filtration [4, Theorem 1.1(i), Theorem 12.1, and §12.2].¹ As in Subsection 5.4, crystalline Frobenius is the pullback of the special

¹For the proper formal model and the algebraic/analytic cohomology identifications, see [7, §7.2, p. 161] and [34, Theorems 9.1 and 9.3], read with [35]. Passage from the algebraic closure to its completion uses [36, Tag 0DDG]. The étale comparisons are applied with $\mathbb{Z}/q^n$ -coefficients, then passed to inverse limits and rationalized.

fiber's q -power endomorphism: Witt Frobenius on $W(\mathbb{F}_q) = \mathbb{Z}_q$ is the identity [36, Tags 07N3 and 07N5]. Crystalline representations give weakly admissible filtered modules [9, Theorem 9.3.4]. The degree-one Hodge filtration has only indices 0 and 1, and functoriality gives the assertion about extending endomorphisms.

The extension by L changes coefficients, not the geometric base field. With Frobenius acting trivially on this coefficient factor, the underlying $\mathbb{Q}_q$ -filtered Frobenius module

$$(D_A \otimes_{\mathbb{Q}_q} L, \varphi_A \otimes 1, F^i D_A \otimes_{\mathbb{Q}_q} L)$$

is a direct sum of $[L : \mathbb{Q}_q]$ copies of D_A , and is weakly admissible. Since p and $1 - p$ both preserve the filtration and commute with Frobenius, their images split it as filtered Frobenius modules. Weak admissibility gives $t_H \leq t_N$ on each image. The Hodge and Newton numbers are additive, and are equal for the whole module. They are therefore equal on each image separately.

Let $m = \dim_{\mathbb{Q}_q} p(D_A \otimes L)$. A pure slope t gives $t_N = tm$ after restriction of scalars: all $\mathbb{Q}_q$ -conjugates of a q -adic algebraic number have the same valuation, including when $L/\mathbb{Q}_q$ is ramified. On the other hand,

$$t_H = \dim_{\mathbb{Q}_q} F^1(p(D_A \otimes L)),$$

because the only filtration indices are 0 and 1. Equality of these two numbers proves the assertion. The general coefficient formalism is also described in [17, Proposition 2.17(ii) and Corollary 2.22]. $\square$

The requirement on the idempotent is essential. For example, $\varphi = \text{diag}(1, q)$ on $\mathbb{Q}_q e_0 \oplus \mathbb{Q}_q e_1$, with only filtration indices 0, 1 and $F^1 = \langle e_0 + e_1 \rangle$, is weakly admissible, but its slope-one line has induced $F^1 = 0$. Its slope projector does not preserve the filtration. The Hecke projectors do preserve it, because they are formed from extending algebraic endomorphisms.

We can therefore obtain a pure Hodge type by proving that a full generalized Hecke space has slope zero or one. The finite-order central relation controls the common scalar in a full extension of the active projective parameter, and the Frobenius polynomial then determines the slope.

Fix an active character η with projective parameter

$$\{q^{1/2}, q^{-1/2}, e'\},$$

where $e' = e$ is supplied by Proposition 6.3, or $e' = \pm \sigma e$ is the parameter of a scalar-conjugate character. Thus e' is algebraic, all its complex conjugates have modulus one, and $v_i(e') \in \{-3/2, 3/2\}$. For a full character χ extending η , its active GL_3 parameter is a scalar multiple of this triple. Absorb the numerical value of C_0 into that scalar and write the resulting triple as

$$(\alpha_1, \alpha_2, \alpha_3) = b(q^{1/2}, q^{-1/2}, e'). \quad (6.7)$$

The determinant coordinate is the character value of $D_3 = C_0^3 T_3$. Lemma 6.4 therefore gives

$$b^3 e' \text{ is a root of unity.} \quad (6.8)$$

In particular b is algebraic and every complex conjugate of b has modulus one.

Write e_i for the elementary symmetric polynomials in three variables. The raw minuscule double cosets, with the integral subgroup of volume one, have Satake transforms qe_1, qe_2, e_3 ; this is [14, Equation (3.14) and §6] for GL_3 . Since $D_i = C_0^i T_i$, absorption of the C_0 -value into b gives

$$\chi(D_1) = qe_1(\alpha), \quad \chi(D_2) = qe_2(\alpha), \quad \chi(D_3) = e_3(\alpha). \quad (6.9)$$

These are numerical character values. On H_χ the operators may still have nilpotent parts.

Proposition 6.6. *Let η be an active numerical character with projective parameter $\{q^{1/2}, q^{-1/2}, e'\}$ supplied by Proposition 6.3 or by scalar conjugation, and put*

$$t_\eta = \frac{1}{2} - \frac{1}{3}v_i(e') \in \{0, 1\}.$$

For every full numerical character χ extending η , the space $H_\chi \otimes_{\overline{\mathbb{Q}}} \mathbb{C}$ is contained in $H^{1,0}(A_S)$ if $t_\eta = 1$, and in $H^{0,1}(A_S)$ if $t_\eta = 0$. Consequently $H_\eta^{\text{act}} \otimes_{\overline{\mathbb{Q}}} \mathbb{C}$ has the same pure Hodge type.

Proof. Fix a full character χ extending η ; if there is none, the assertion is vacuous. We first use Weil weight to select the Frobenius root. We then apply Lemma 6.5 and compare the resulting filtration ranks with those on the complex Betti realization.

At the fixed completion $K_v = \mathbb{Q}_q$, write

$$D_S = H_{\text{dR}}^1((A_S)_{K_v}/\mathbb{Q}_q)$$

and let φ be its crystalline Frobenius. If Φ is the q -power Frobenius endomorphism of $\tilde{A}_S/\mathbb{F}_q$ used in Proposition 5.5, then $\varphi = \Phi^*$ under crystalline comparison. The Hecke endomorphisms extend over $\mathbb{Z}_q$; their reductions commute with Φ , so their pullbacks commute with φ .

By additivity of pullback on abelian degree-one cohomology, the integral identity of Proposition 5.5 gives

$$\varphi^3 - D_1^* \varphi^2 + q D_2^* \varphi - q^3 D_3^* = 0. \quad (6.10)$$

Pullback reverses composition, but all operators in this formula commute.

Choose a prime $\ell \neq q$ and put $V_\ell \tilde{A}_S = T_\ell \tilde{A}_S \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$. Weil's theorem for abelian varieties over $\mathbb{F}_q$ gives

$$P_\Phi(X) = \det(X - \Phi \mid V_\ell \tilde{A}_S) \in \mathbb{Z}[X],$$

all of whose complex roots have absolute value $q^{1/2}$. The prime-to- q Tate-module action on endomorphisms is faithful. Cayley–Hamilton therefore gives $P_\Phi(\Phi) = 0$ in the endomorphism ring, and degree-one additivity gives $P_\Phi(\varphi) = 0$. These statements hold for an arbitrary abelian variety over $\mathbb{F}_q$; see [24, Lemma 12.2, Proposition 12.9, and Theorem 19.1(c) with its proof]. No CM or ordinary hypothesis on $\tilde{A}_S$ is involved. This uses the q -power endomorphism and its pullback, not a reciprocal Galois-Frobenius convention.

Transport the algebraic coefficients of p_χ and χ by ι . On the corresponding generalized summand of $D_S \otimes_{\mathbb{Q}_q} \overline{\mathbb{Q}_q}$, the commuting Hecke maps and φ can be simultaneously triangularized. Equations (6.10) and (6.9) imply that every diagonal Frobenius eigenvalue is a root of

$$X^3 - qe_1(\alpha)X^2 + q^2e_2(\alpha)X - q^3e_3(\alpha) = \prod_{j=1}^3 (X - q\alpha_j).$$

Thus the candidates, before applying ι , are

$$bq^{3/2}, \quad bq^{1/2}, \quad qbe'. \quad (6.11)$$

Each is algebraic. If its image under ι is a root of the rational polynomial P_Φ , the algebraic number itself is a root of P_Φ . We may therefore apply the complex absolute-value condition. The three absolute values are respectively

$$q^{3/2}, \quad q^{1/2}, \quad q,$$

because every complex conjugate of b and e' has modulus one. Only $bq^{1/2}$ remains.

The weight condition has now selected one Newton slope on the entire generalized summand. Equation (6.8) gives

$$v_i(bq^{1/2}) = \frac{1}{2} - \frac{1}{3}v_i(e') = t_\eta. \quad (6.12)$$

This conclusion concerns every Frobenius eigenvalue; Jordan blocks do not change it. Choose a finite extension $L/\mathbb{Q}_q$ containing the coefficients of the transported idempotent and numerical character. The idempotent p_χ is a polynomial in extending Hecke maps, so it preserves the Hodge filtration and commutes with φ . Lemma 6.5 makes its filtration pure of index t_η .

It remains to relate this filtered rank to the Hodge type of the same complex generalized space. Put

$$V_{\text{dR}} = H_{\text{dR}}^1(A_S/K).$$

Its filtration and Hecke operators are defined over the fixed field K . For a numerical character χ , choose a polynomial expression for p_χ , and let $k_\chi \subset \overline{\mathbb{Q}}$ be the actual compositum of K and its algebraic coefficients. Evaluating that polynomial in the fixed operators gives an idempotent on $V_{\text{dR}} \otimes_K k_\chi$. Its induced filtration is its projected filtration, since the idempotent preserves $F^\bullet$. The dimensions of its image and of this filtration are ranks of matrices over k_χ ; they are unchanged under the restriction of the fixed embedding ι to k_χ .

Over the given embedding $K \subset \mathbb{C}$, Betti–de Rham comparison intertwines the same operators and polynomial. It identifies these ranks with the filtration ranks on $H_\chi \otimes_{\overline{\mathbb{Q}}} \mathbb{C}$. For a scalar-conjugate numerical character $\sigma\chi$, repeat this argument with $p_{\sigma\chi}$ in the same K -defined operators. On Betti cohomology their rational matrices give $\sigma H_\chi = H_{\sigma\chi}$. We have therefore computed the type of the scalar-conjugated Betti space on the same complex variety. Each conjugate projector is evaluated separately in the fixed K -defined operators. This works even when K intersects the character field, since it uses ranks over their actual compositum.

Finally the complex Hecke operators preserve both $H^{1,0}$ and $H^{0,1}$. Hence a full projected F^1 means that the whole generalized space is holomorphic, and zero projected F^1 means that it is antiholomorphic. Equation (6.12) depends only on η , so summing all full extensions gives the assertion for H_η^{act} . $\square$

6.3 Rational CM sources for the theta classes

The preceding proposition determines the type of every vector in each scalar-conjugate active Hecke space. We can therefore apply scalar descent to maps with arbitrary algebraic Betti vectors in these spaces. This will prove the rational-map span asserted in Theorem 5.1.

Proof of Theorem 5.1. Use the Albanese identification $H^1(A_S, \mathbb{Q}) \simeq H^1(S, \mathbb{Q})$. Extend a theta class from S_0 by zero on the other components of S . This is compatible with the active algebra, because Lemma 6.1 makes that algebra component-preserving. The class is holomorphic by Proposition 4.5. Decompose it into the active generalized character spaces of Equation (6.6). For each nonzero component, let η be its active character and e its parameter from Proposition 6.3. For every $\sigma \in \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, put $g = \sigma|_E$. That proposition and Proposition 6.6 give

$$v_i(\sigma e) = -\frac{3}{2}v(g), \quad t_{\sigma\eta} = \frac{1}{2} - \frac{1}{3}v_i(\sigma e) = \frac{1+v(g)}{2}. \quad (6.13)$$

The possible sign in the conjugate parameter $e' = \pm\sigma e$ has valuation zero. Consequently the entire active space for $\sigma\eta$ is of type $(1,0)$ when $v(g) = 1$ and of type $(0,1)$ when $v(g) = -1$. This applies to all of its vectors, not just to theta vectors. At the identity it gives type $(1,0)$, in agreement with $v(1) = 1$ and the holomorphy of the theta class.

Each active space is defined over $\overline{\mathbb{Q}}$ in the rational Betti structure, because its idempotent has algebraic coefficients. Choose $0 \neq x \in H_{\eta}^{\text{act}}$ and $0 \neq u \in U(v)_1(\overline{\mathbb{Q}})$. Define a $\overline{\mathbb{Q}}$ -linear map

$$h : U(v)_{\overline{\mathbb{Q}}} \longrightarrow H^1(S, \overline{\mathbb{Q}})$$

by $h(u) = x$ and by setting it equal to zero on every other embedding line of $U(v)_{\overline{\mathbb{Q}}}$. This is an element of

$$\text{Hom}_{\mathbb{Q}}(U(v), H^1(S, \mathbb{Q})) \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}.$$

Scalar conjugation by σ moves its nonzero source line to $U(v)_{\sigma}$ and its target vector to $\sigma x \in H_{\sigma\eta}^{\text{act}}$. The source line has the type prescribed by $v(g)$, and Equation (6.13) gives the same type for the target. All other source lines map to zero. Hence every scalar conjugate of h is a Hodge map.

Lemma 2.1, applied to the complex-linear subspace of Hodge maps, expresses h as a $\overline{\mathbb{Q}}$ -linear combination of rational Hodge maps. Algebraic-Betti vectors span each complex active space, so taking complex spans covers every active component of the original theta class, including classes with transcendental coefficients. Restriction from S to S_0 is a rational Hodge map and gives Equation (5.1).

For a finite collection of inputs, all level choices are made before the auxiliary prime is chosen, and only finitely many primes are excluded. The preceding argument therefore applies to that collection at one sufficiently deep level. If $\pi : S'_0 \rightarrow S_0$ is a deeper principal cover, then π^* is a rational Hodge map and the theta classes pull back to their classes on S'_0 . Applying π^* to Equation (5.1) proves persistence. $\square$

By Lemma 2.2, each rational map in Equation (5.1) is, up to a rational multiple, pullback along a morphism $S_0 \rightarrow B(v)$.

Corollary 6.7 (Finite-family CM extraction). *Fix E, V, D, Λ and the principal lattice levels of Section 5. Let $v_1, \dots, v_m : G_E \rightarrow \{1, -1\}$ be odd with $v_j(1) = 1$. For each j , choose a finite family of finite adelic Schwartz inputs in the multiplicity-line construction $d_j(\tau) = v_j(\tau^{-1})$. Let X_{base} be a principal quotient at a torsion-free level in this framework. There is a common deeper principal cover $Y \rightarrow X_{\text{base}}$, with Y a smooth connected projective surface, on which every associated theta one-form class from the j -th family belongs to*

$$\text{span}_{\mathbb{C}} \left\{ f(U(v_j)_1) : f \in \text{Hom}_{\text{Hdg}}(U(v_j), H^1(Y, \mathbb{Q})) \right\}.$$

The maps are rational Hodge maps, with their images viewed after extension to $\mathbb{C}$. The conclusion persists after pullback to every deeper principal level. The auxiliary primes used for different v_j may differ.

Proof. Apply Theorem 5.1 to each finite input family. Choose a principal level contained in the resulting finitely many principal levels and in the level of X_{base} , and deepen it until Lemma 5.2 applies. Its quotient Y is smooth, connected, and projective. The reference moduli construction depends on V , not on the multiplicity line, so it applies to this common level. The persistence in Theorem 5.1 pulls all the rational-map spans to Y and then to every deeper principal level. Each application chooses its auxiliary prime only after its own finite family and level have been fixed; the primes need not agree. $\square$

7 Completion of the CM proof

The theta construction gives a nonzero period on a compact ball quotient. The finite-family extraction corollary expresses the one-form classes in that period through rational maps from the prescribed CM modules, after passage to sufficiently deep levels. We put these statements on one projective surface and apply the criterion of Section 3.

Proof of Theorem 2.5. By Lemma 2.7, it suffices to treat the case $s \geq 2$ and $v_i(1) = 1$ for all four functions. Use the common Galois CM field E , the active embedding $\mu = 1$, and the line signs

$$d_i(\tau) = v_i(\tau^{-1}).$$

They satisfy $d_1 + d_2 = d_3 + d_4$ and $d_i(\mu) = 1$. Choose the rank-three Hermitian space V , the ball D , and the rational orthogonal decomposition $W = W_1 \perp W_2$ prescribed by d_1, d_2 , as in Section 4.

Proposition 4.6(ii) supplies a total finite theta input whose form for this decomposition is nonzero. With that input fixed, Corollary 4.7 supplies a second rational orthogonal decomposition and a nonzero mixed period. Write X' for the individual-factor cover constructed there, and ξ_1, ξ_2 for the two pulled-back endpoint forms. On this common compact principal quotient they have finite expansions

$$\xi_1 = \sum_{a \in I} c_a \Theta_1(\phi_{1a}) \wedge \Theta_2(\phi_{2a}), \quad \xi_2 = \sum_{b \in J} c'_b \Theta_3(\phi_{3b}) \wedge \Theta_4(\phi_{4b}),$$

where the constants account for the chosen constant-line trivializations, and

$$\int_{X'} \xi_1 \wedge \overline{\xi_2} \neq 0.$$

All four families of individual inputs here are finite. The quotient and these expansions may be pulled to any deeper principal level; the integral is then multiplied by the degree of the cover.

Apply Corollary 6.7 to these four families, with $X_{\text{base}} = X'$ and the fixed E, V, D, Λ . It gives a common smooth connected projective principal cover $\pi : Y \rightarrow X'$ on which all four $U(v_i)_1$ -span conclusions hold; the auxiliary primes may differ.

Denote the pulled-back one-form classes by

$$\vartheta_{ia} = [\pi^* \Theta_i(\phi_{ia})] \in H^1(Y, \mathbb{C}),$$

with $a \in I$ for $i = 1, 2$ and $a \in J$ for $i = 3, 4$. These classes are represented by holomorphic one-forms. Their cup products compute the integrals of the wedge products, and the covering formula gives

$$\begin{aligned} 0 &\neq \int_Y \pi^* \xi_1 \wedge \overline{\pi^* \xi_2} \\ &= \sum_{a \in I, b \in J} c_a \overline{c'_b} \langle \vartheta_{1a} \cup \vartheta_{2a} \cup \overline{\vartheta_{3b}} \cup \overline{\vartheta_{4b}}, [Y] \rangle. \end{aligned} \tag{7.1}$$

Choose a, b for which the displayed cup-product period is nonzero, and write its four holomorphic factors as ϑ_i .

Corollary 6.7 gives

$$\vartheta_i \in \text{span}_{\mathbb{C}} \{h(U(v_i)_1) : h \in \text{Hom}_{\text{Hdg}}(U(v_i), H^1(Y, \mathbb{Q}))\}, \quad 1 \leq i \leq 4.$$

The maps are rational, so they commute with scalar complex conjugation, which sends $U(v_i)_1$ to $U(v_i)_c$. Consequently

$$\overline{\vartheta_i} \in \text{span}_{\mathbb{C}} \{h(U(v_i)_c) : h \in \text{Hom}_{\text{Hdg}}(U(v_i), H^1(Y, \mathbb{Q}))\}, \quad i = 3, 4.$$

The four classes $\vartheta_1, \vartheta_2, \overline{\vartheta_3}, \overline{\vartheta_4}$ thus satisfy Corollary 3.2 on the same projective surface Y , with the source labels $(1, 1, c, c)$ and the nonzero period selected from Equation (7.1). The normalized switch line is therefore algebraic. Lemma 2.7 returns every switch to this case or to a product of algebraic two-factor classes, and proves the theorem. $\square$

Proof of Theorem 1.1. The four-factor switch theorem is now proved for every Galois CM field. Proposition 2.6 consequently places every rational Hodge class on a CM abelian variety in its rational cycle span. Its final scalar descent takes place on the original variety. Finite products and powers are CM, so the same conclusion covers them. $\square$

8 Generalized Hodge and arithmetic consequences

Theorem 1.1 applies to products of CM abelian varieties and therefore algebraizes rational Hodge correspondences between them. We use this to obtain the algebraic supports predicted by the generalized Hodge conjecture, following Abdulali's effective-realization argument [1, Lemma 1 and Theorems 3–4]. Hazama had earlier reduced this implication to codimension-two Hodge classes on a special family of CM abelian varieties [16, Theorem 8.3]. We then apply Milne's Hodge-to-Tate and polarization theorems to obtain the Tate and Hodge standard conjectures for abelian varieties. A final application uses CM transport to algebraize specialized Hodge classes on arbitrary good-reduction abelian varieties.

8.1 The generalized Hodge conjecture

A pure rational Hodge structure is *effective* if all its nonzero Hodge components have nonnegative indices. Recall that $V(r)^{p,q} = V^{p+r,q+r}$, so the twist $V(r)$ has weight $\text{wt}(V) - 2r$. The Hodge circle of a pure Hodge structure V acts on $V^{p,q}$ by z^{p-q} , for $|z| = 1$. Let $J(V)$ be the smallest algebraic subgroup of $\text{GL}(V)$ over $\mathbb{Q}$ whose real points contain that circle. A polarizable V is of *CM type* if $J(V)$ is a torus, with the trivial group allowed. Every Hodge substructure of the cohomology of a CM abelian variety is of CM type: its Hodge-circle group is an image of the torus acting on degree-one cohomology.

For a smooth projective complex variety X , the geometric coniveau filtration is

$$N^r H^k(X, \mathbb{Q}) = \sum_{\substack{Z \subset X \text{ closed} \\ \text{codim}_X Z \geq r}} \ker(H^k(X, \mathbb{Q}) \longrightarrow H^k(X \setminus Z, \mathbb{Q})).$$

Grothendieck's corrected generalized Hodge conjecture says that a rational Hodge substructure $W \subset H^k(X, \mathbb{Q})$ lies in $N^r H^k(X, \mathbb{Q})$ whenever $W(r)$ is effective; see [15, §2] and [16, §2]. Equivalently, one closed subset of codimension at least r supports all of W . Indeed, finite dimensionality permits taking a finite union of the supports needed for a basis of W .

We will obtain such a support by algebraizing a Hodge map from an auxiliary CM abelian variety. To get the required codimension, that variety must realize an effective twist of W in cohomology of the same weight; the twisting index will determine the codimension of the resulting correspondence. The next lemma constructs it by expressing the Hodge indices at each embedding as sums of weight-one indices.

Lemma 8.1 (Effective realization of a simple CM Hodge structure). *Let W be a nonzero simple polarizable rational Hodge structure of CM type and weight k . Put*

$$r_0 = \min\{p : W^{p,k-p} \neq 0\}, \quad n = k - 2r_0.$$

Then $n \geq 0$, and there is a CM abelian variety B , possibly a point, and a surjective rational Hodge map

$$H^n(B, \mathbb{Q})(-r_0) \longrightarrow W.$$

Proof. Set $U = W(r_0)$. Hodge symmetry shows that U is effective of weight $n \geq 0$, with smallest Hodge index zero. If $n = 0$, every type of U is $(0, 0)$. Simplicity then gives $U \simeq \mathbb{Q}$, and we take B to be a point.

Suppose $n > 0$. We first show that U is a line over a CM field; we will then decompose its Hodge type into weight-one CM types. Write $T = J(U)$. Let E_0 be the $\mathbb{Q}$ -subalgebra of $\text{End}_{\mathbb{Q}}(U)$ generated by the image of T . After splitting the torus, its action is diagonal by characters, so E_0 is a commutative finite étale algebra. Its field idempotents and every subspace linear over one field factor are T -stable. Simplicity of U therefore forces E_0 to be a field E and $\dim_E U = 1$.

We check that E is a CM field. Let the Weil operator C_U act as i^{p-q} on $U^{p,q}$. By polarizability there is a rational Hodge pairing ψ , with the conventional weight-dependent sign absorbed in its definition, for which

$$q(x, y) = \psi(C_U x, y)$$

is positive definite on $U_{\mathbb{R}}$. The adjoint $\dagger$ for ψ sends $t \in T$ to t^{-1} , hence preserves E . Every element of E commutes with C_U , so $\dagger$ is also the adjoint for q . On the real finite étale algebra $E_{\mathbb{R}}$, it must fix each primitive idempotent: if an idempotent e were sent to a different one, then $e^{\dagger}e = 0$, contrary to $q(ex, ex) = q(x, e^{\dagger}ex) > 0$ for some x . On a complex factor it must act by complex conjugation, since the identity would give $i^{\dagger}i = -1$, again contrary to positivity. Thus if $\dagger$ is trivial on E , the field is totally real. If it is nontrivial, a real embedding would force $\dagger$ to be trivial on E , so there are no real embeddings; the fixed algebra in $E_{\mathbb{R}}$ is then a product of copies of $\mathbb{R}$. Its fixed field is therefore totally real, and E is CM.

The totally real case cannot occur here. The Hodge circle acts through $E_{\mathbb{R}}^{\times}$; its image in each real scalar factor is a compact connected subgroup of $\mathbb{R}^{\times}$, hence is trivial. All Hodge types of U would then have equal indices. Since their smallest index is zero, this would force $n = 0$. Consequently E is CM, and $\dagger$ is its complex conjugation.

Let $\Sigma = \text{Hom}(E, \mathbb{C})$, and write $\bar{\sigma}$ for the conjugate of σ . Since E acts by Hodge endomorphisms and U is an E -line,

$$U_{\mathbb{C}} = \bigoplus_{\sigma \in \Sigma} U_{\sigma}, \quad U_{\sigma} \text{ has type } (a_{\sigma}, n - a_{\sigma}),$$

where $0 \leq a_{\sigma} \leq n$ and $a_{\sigma} + a_{\bar{\sigma}} = n$. Choose n CM types $\Phi_1, \dots, \Phi_n$ of E with

$$\#\{j : \sigma \in \Phi_j\} = a_{\sigma}.$$

This is possible independently for each conjugate pair: put σ into any a_{σ} lists and $\bar{\sigma}$ into the remaining lists.

For each j , give the rational vector space $H_j = E$ the weight-one Hodge structure whose σ -line has type $(1, 0)$ exactly when $\sigma \in \Phi_j$. The polarization and Riemann construction of Section 2, surrounding Equation (2.2), applies to any CM field, with no Galois hypothesis. It gives a CM abelian variety B_j with $H^1(B_j, \mathbb{Q}) \simeq H_j$.

Identify U and the H_j with E as E -lines. The rational multiplication map

$$E^{\otimes \mathbb{Q}^n} \longrightarrow E, \quad x_1 \otimes \cdots \otimes x_n \longmapsto x_1 \cdots x_n$$

is surjective. Over $\mathbb{C}$ it kills tensors with unequal embedding labels, while the tensor with all labels equal to σ has type $(a_{\sigma}, n - a_{\sigma})$ and maps to U_{σ} . It is therefore a Hodge map to U . For $B = \prod_{j=1}^n B_j$, its source is the rational Künneth summand $\bigotimes_j H^1(B_j, \mathbb{Q})$ of $H^n(B, \mathbb{Q})$. Extending by zero on the other summands and twisting by $(-r_0)$ proves the lemma. $\square$

Corollary 8.2 (Generalized Hodge conjecture for CM abelian varieties). *Let A be a complex CM abelian variety. For integers $k, r \geq 0$ and a rational Hodge substructure $W \subset H^k(A, \mathbb{Q})$ such that $W(r)$ is effective, there is a closed algebraic subset $Z \subset A$ of codimension at least r such that*

$$W \subset \ker(H^k(A, \mathbb{Q}) \longrightarrow H^k(A \setminus Z, \mathbb{Q})).$$

This includes every finite product of CM abelian varieties and every power of such a product.

Proof. The zero substructure permits $Z = \emptyset$. Otherwise, semisimplicity of polarizable rational Hodge structures reduces the claim to a simple summand W , after which we take the finite union of the resulting supports. Let r_0 be its smallest Hodge index. Then $r_0 \geq r$. Also $r_0 \leq k/2 \leq \dim A$, since $W \neq 0$. Put $n = k - 2r_0$, take B from Lemma 8.1, and write $b = \dim B$. Composing the map of that lemma with $W \hookrightarrow H^k(A, \mathbb{Q})$ gives

$$f : H^n(B, \mathbb{Q})(-r_0) \longrightarrow H^k(A, \mathbb{Q}), \quad \text{im}(f) = W.$$

By Poincaré duality, f corresponds to a rational Hodge class of type $(b + r_0, b + r_0)$ in

$$H^{2b-n}(B, \mathbb{Q}) \otimes H^k(A, \mathbb{Q}) \subset H^{2b+2r_0}(B \times A, \mathbb{Q}).$$

The product $B \times A$ is CM. Theorem 1.1 therefore represents this exact class by $\Gamma \in \text{CH}^{b+r_0}(B \times A)_{\mathbb{Q}}$, whose action in degree n is f .

Choose a finite rational cycle representing Γ , and let Z be the projection of its support to A . Every component of that cycle has dimension

$$\dim(B \times A) - (b + r_0) = \dim A - r_0.$$

The projection is proper, so Z is closed and $\text{codim}_A Z \geq r_0 \geq r$. The correspondence restricts to zero on $B \times (A \setminus Z)$; functoriality of its cohomological action therefore makes its image W restrict to zero on $A \setminus Z$. $\square$

8.2 The Tate conjecture over finite fields

Let A be an abelian variety over $\mathbb{F}_q$, put $\bar{A} = A \times_{\mathbb{F}_q} \overline{\mathbb{F}_q}$, and write $G_q = \text{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q)$. For a prime $\ell \neq \text{char}(\mathbb{F}_q)$, the rational Tate conjecture in codimension r asks whether cycles over $\mathbb{F}_q$ span $H_{\text{ét}}^{2r}(\bar{A}, \mathbb{Q}_{\ell}(r))^{G_q}$.

Corollary 8.3 (Tate conjecture for abelian varieties over finite fields). *For every abelian variety $A/\mathbb{F}_q$, every integer $r \geq 0$, and every prime $\ell \neq \text{char}(\mathbb{F}_q)$, the cycle-class map*

$$\text{CH}^r(A) \otimes_{\mathbb{Z}} \mathbb{Q}_{\ell} \longrightarrow H_{\text{ét}}^{2r}(\bar{A}, \mathbb{Q}_{\ell}(r))^{G_q}$$

is surjective. Numerical equivalence and ℓ -adic homological equivalence coincide on rational algebraic cycles on $\bar{A}$.

Proof. Apply Milne's Theorem 7.1 and his formulation (0.1) [25], with the correction in [22]: the cofinal auxiliary Galois CM fields are enlarged to contain a real subfield of degree > 2 inert at the residue prime. The theorem's hypothesis is supplied by Theorem 1.1. It gives the stated equality of equivalence relations and algebraic spanning of the geometric Tate classes: those fixed over some finite field of definition.

Let α lie in the displayed G_q -invariant space. The geometric conclusion writes α as a finite $\mathbb{Q}_{\ell}$ -linear combination of cycle classes. All the cycles in this expression descend to $A_{\mathbb{F}_{q^m}}$ for one m ; let

$z_m \in \mathrm{CH}^r(A_{\mathbb{F}_{q^m}}) \otimes_{\mathbb{Z}} \mathbb{Q}_\ell$ denote the resulting combination. For the finite projection $\pi_m : A_{\mathbb{F}_{q^m}} \rightarrow A$, proper pushforward and base change for cycle classes give

$$\mathrm{cl}_\ell \left(\frac{1}{m} (\pi_m)_* z_m \right) = \frac{1}{m} \sum_{\sigma \in \mathrm{Gal}(\mathbb{F}_{q^m}/\mathbb{F}_q)} \sigma(\alpha) = \alpha.$$

Thus $m^{-1}(\pi_m)_* z_m$ is the required class over $\mathbb{F}_q$. Division by m is valid in $\mathbb{Q}_\ell$, even when m is divisible by ℓ or by the characteristic. $\square$

Milne's implication compares CM and finite-field abelian motives. It does not assert that each Tate class lifts to a Hodge class on a single CM lifting, even up to isogeny [25, Introduction, Remark (a)].

8.3 The Hodge standard conjecture

Let A be an abelian variety of dimension g over an algebraically closed field K , and let $A_{\mathrm{num}}^r(A)_{\mathbb{Q}}$ be the space of rational codimension- r cycles modulo numerical equivalence. A *rational ample divisor class* is an $h \in A_{\mathrm{num}}^1(A)_{\mathbb{Q}}$ for which some positive integral multiple is the numerical class of an ample Cartier divisor. For integers $0 \leq r \leq g/2$, put

$$P_h^r(A) = \ker \left(h^{g-2r+1} : A_{\mathrm{num}}^r(A)_{\mathbb{Q}} \longrightarrow A_{\mathrm{num}}^{g-r+1}(A)_{\mathbb{Q}} \right).$$

Corollary 8.4 (Hodge standard conjecture for abelian varieties). *Let K be an algebraically closed field and A/K an abelian variety of dimension g . For every prime $\ell \neq \mathrm{char}(K)$, numerical and ℓ -adic homological equivalence coincide on rational algebraic cycles on A . For every rational ample divisor class h and every integer $0 \leq r \leq g/2$, the rational symmetric form*

$$P_h^r(A) \times P_h^r(A) \longrightarrow \mathbb{Q}, \quad (x, y) \longmapsto (-1)^r \deg(xy h^{g-2r})$$

is positive definite after extension of scalars to $\mathbb{R}$. Thus the Hodge standard conjecture for ℓ -adic étale cohomology holds for abelian varieties in arbitrary characteristic.

Proof. Apply Milne's Theorem 3.3 [26] using Theorem 1.1. In Milne's proof, the same Hodge hypothesis supplies the compatibility and positivity of the relevant polarizations through Theorem 2.1, read with the author's addendum [23] and the corrected auxiliary-field choice from [22]. Theorem 3.3 gives both the equality of equivalence relations and positivity on the primitive space of cycles modulo homological equivalence. The former identifies that primitive space with the numerical one above.

The case $g = 0$ is immediate. For $g > 0$, choose a positive integer m so that mh is the class of a smooth hyperplane section; clearing denominators, taking a sufficiently large very ample multiple, and Bertini's theorem permit this. Multiplication by the nonzero rational scalar m^{g-2r+1} does not change the primitive kernel, and the form for mh is m^{g-2r} times the form for h . This positive factor preserves positive definiteness. $\square$

8.4 Specialization of Hodge classes

Remark 8.5 (Algebraicity of specialized Hodge classes). Theorem 1.1 also combines with the CM transport of [32, Corollary 3.4] to give a specialization statement for abelian varieties that need not be CM. Fix a prime p and a p -adic place of $\overline{\mathbb{Q}}$. If $A/\overline{\mathbb{Q}}$ has good reduction $A_0/\overline{\mathbb{F}}_p$ at that

place, every rational Hodge class on A specializes to the class of one rational algebraic cycle on A_0 , simultaneously in all prime-to- p realizations and in crystalline cohomology after extension to the completed field at the chosen place. Indeed, the transport writes that specialization as the pullback of a specialized CM Hodge class by a rational homomorphism of special fibers. Theorem 1.1 algebraizes the CM class; specialization and Chow pullback after clearing the homomorphism's denominator give the required cycle. See [32, Corollary 6.4] for the field-of-definition argument and precise realization conventions, including $p = 2$. This application uses the CM transport, not the independent rationality proof of that paper, and is not an input to Theorem 1.1. It asserts algebraicity on the special fiber, not the Hodge conjecture for arbitrary complex abelian varieties.

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